Hypergraph: Modeling Higher-Order Interactions in the Real World

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InHyeok Jeong
School of Computer Science and Engineering
Chung-Ang University
MINDS Lab

Content

- ☐ Introduction
- ☐ Hypergraph Expansions for Message Passing Design
- ☐ Self-Supervised Hypergraph Learning

From pairwise to group-wise

Higher-Order Interactions are Everywhere

- **Co-authorships of Researchers**

Enhancing Hyperedge Prediction with Context-Aware Self-Supervised Learning

Yunyong Ko, Hanghang Tong, *Fellow, IEEE,* and Sang-Wook Kim*, *Senior Member, IEEE*

HNHN: Hypergraph Networks with Hyperedge Neurons

Yihe Dong
Microsoft

Will Sawin
Department of Mathematics
Columbia University

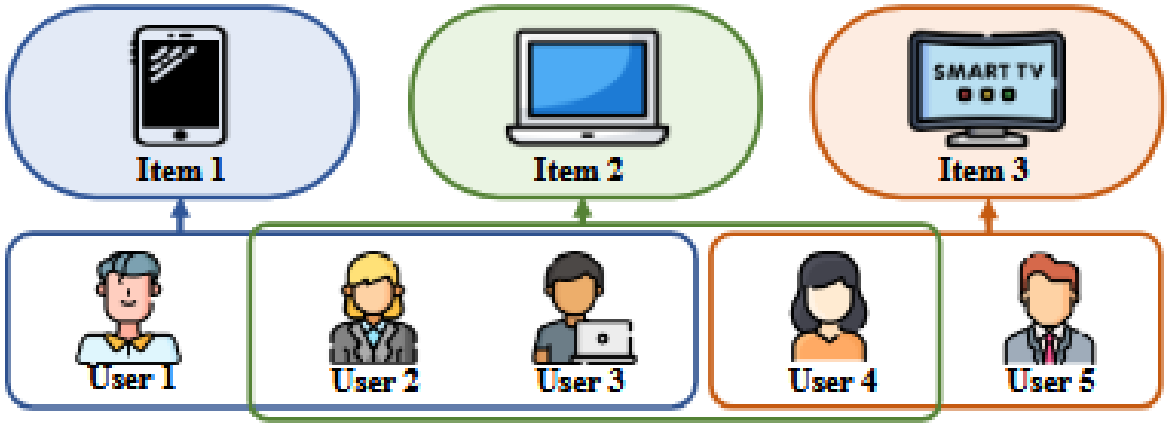
Yoshua Bengio
Mila
Université de Montréal

Ko, Yunyong, Hanghang Tong, and Sang-Wook Kim. "Enhancing hyperedge prediction with context-aware self-supervised learning." *IEEE Transactions on Knowledge and Data Engineering* (2025).

Dong, Yihe, Will Sawin, and Yoshua Bengio. "Hnhn: Hypergraph networks with hyperedge neurons." *arXiv preprint arXiv:2006.12278* (2020).

Higher-Order Interactions are Everywhere

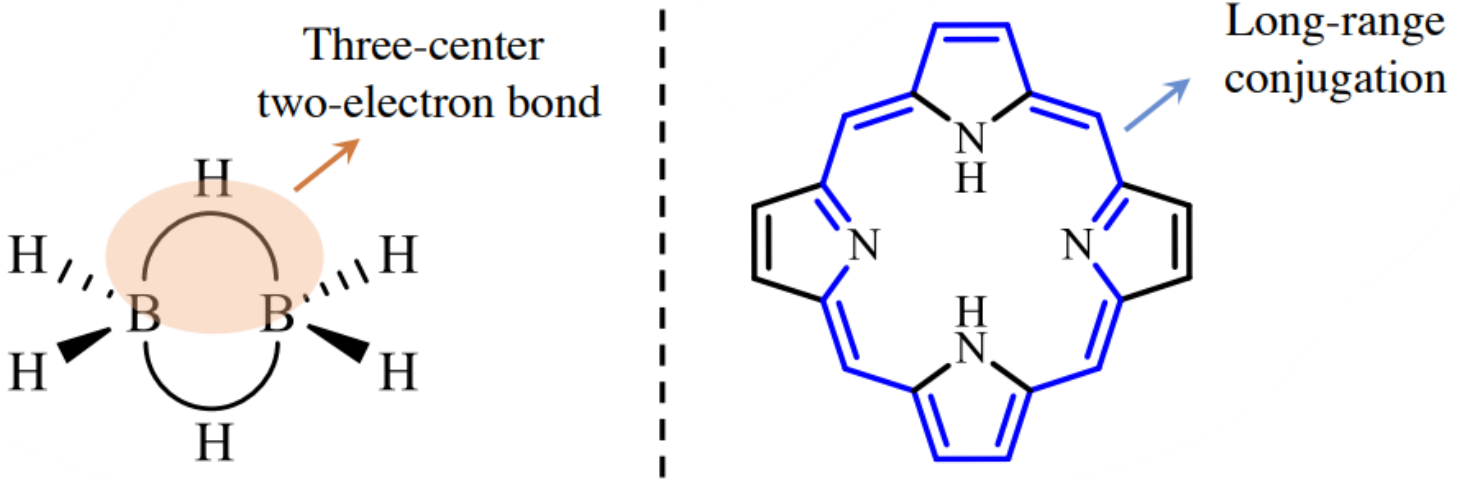
- Co-purchase of items



Ko, Yunyong, Hanghang Tong, and Sang-Wook Kim. "Enhancing hyperedge prediction with context-aware self-supervised learning." *IEEE Transactions on Knowledge and Data Engineering* (2025).

Higher-Order Interactions are Everywhere

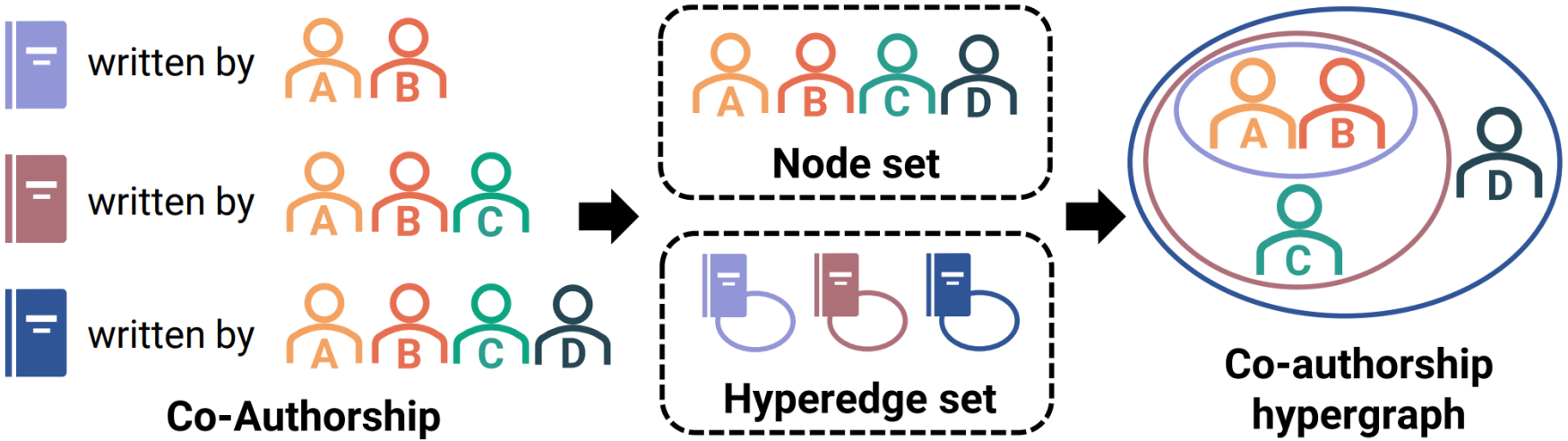
☐ Molecule Structure



Chen, Junwu, and Philippe Schwaller. "Molecular hypergraph neural networks." *The Journal of Chemical Physics* 160.14 (2024).

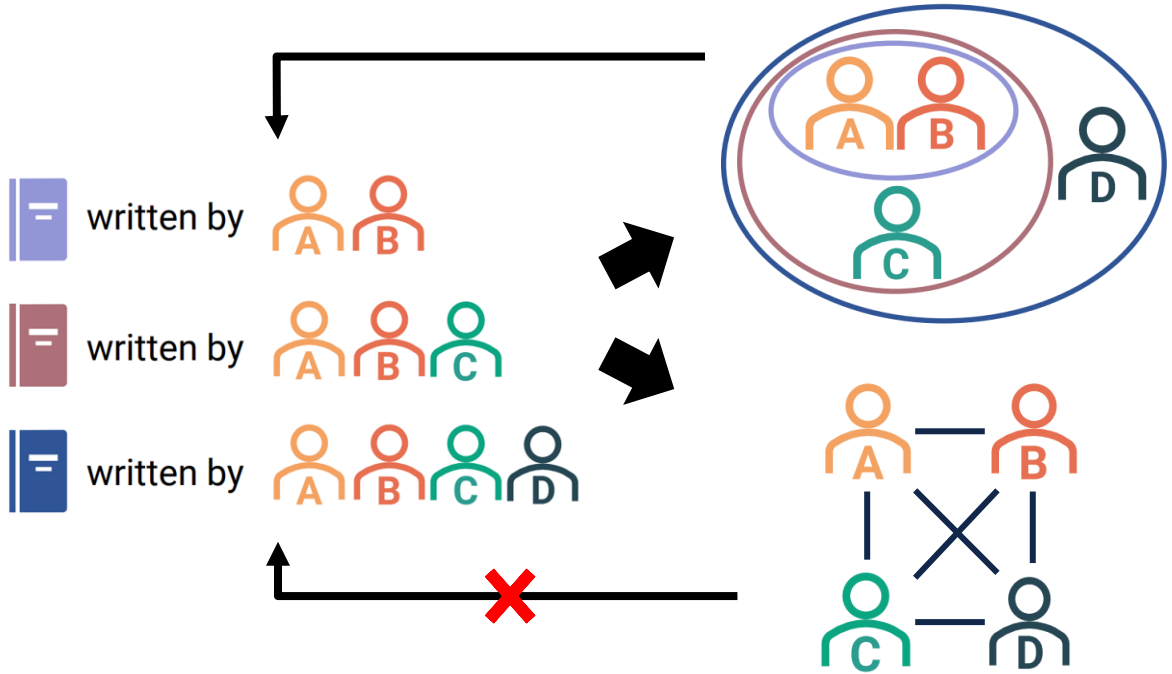
Hypergraphs Model Higher-Order Interactions

- **Hypergraphs can naturally model higher-order interactions as hyperedges**
  - A **hyperedge** is subset of vertices



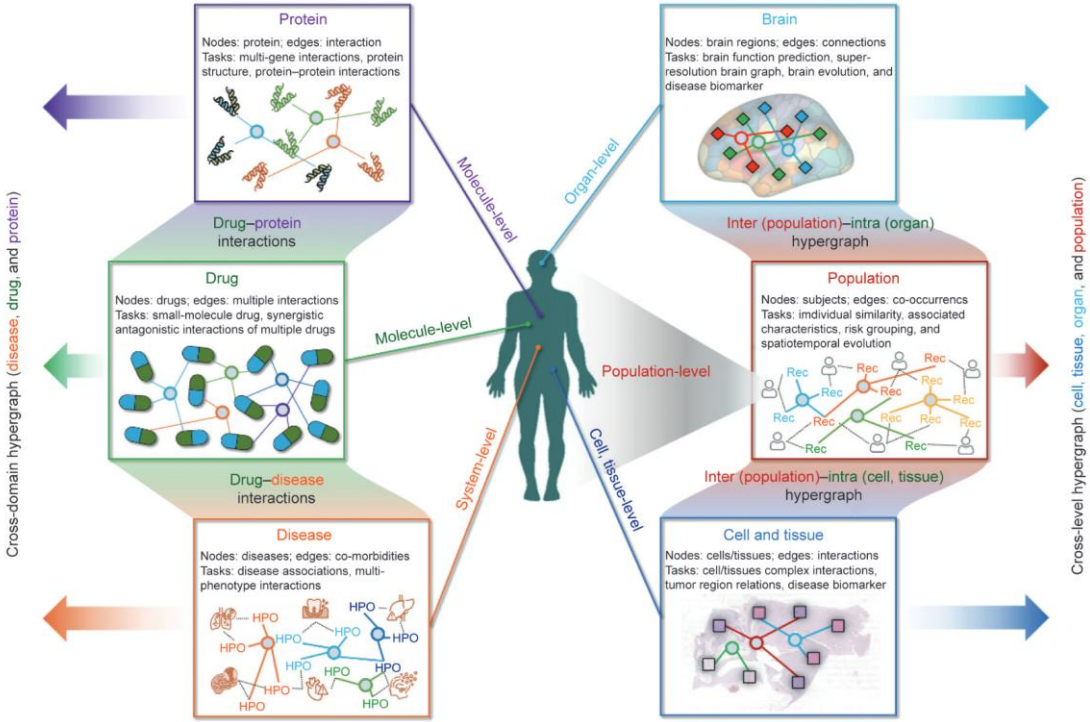
Limitation of Graphs

- **Using graphs can incur information loss**
  - Unable to recover higher-order interactions from simple graphs



Application

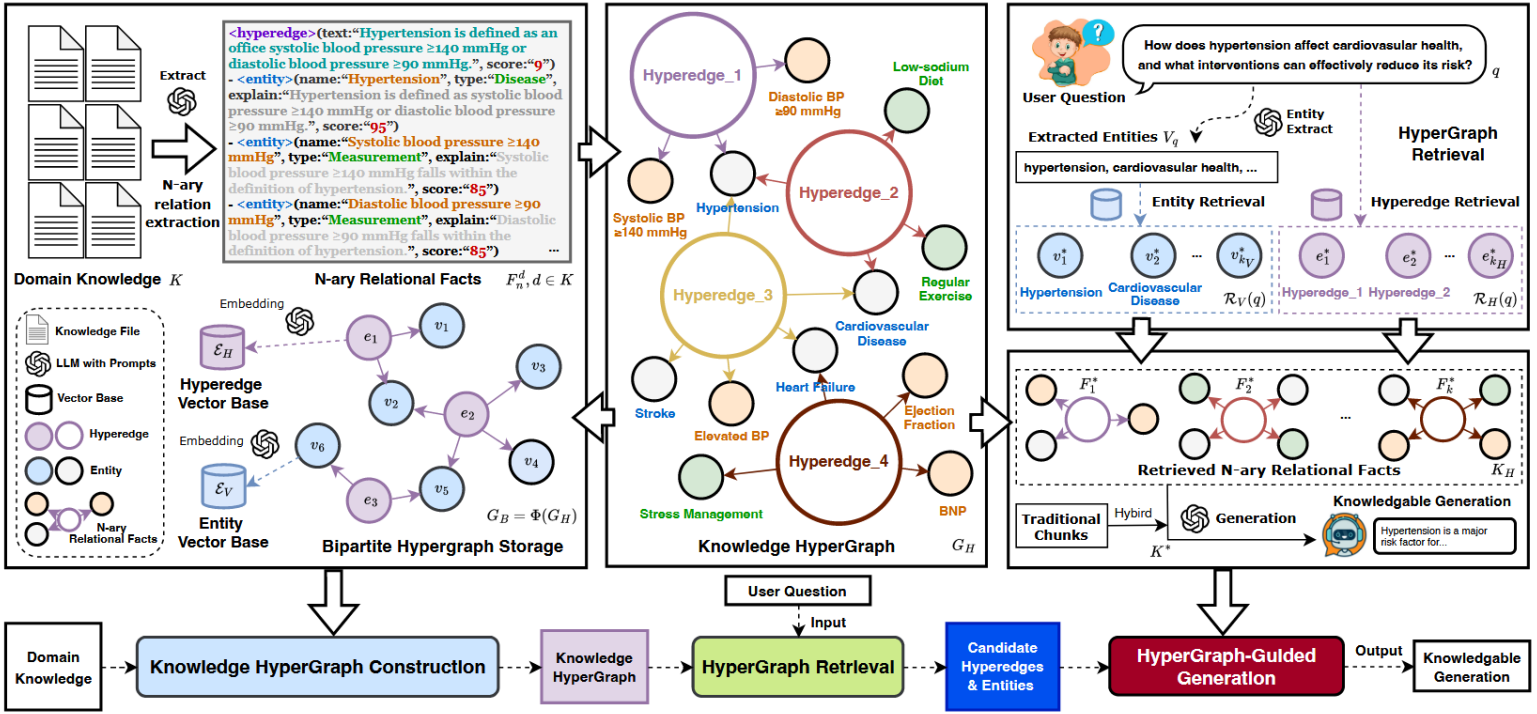
☐ Intelligent Medicine



Gao, Yue, et al. "Hypergraph computation." *Engineering* 40 (2024): 188-201.

Application

☐ Information Retrieval

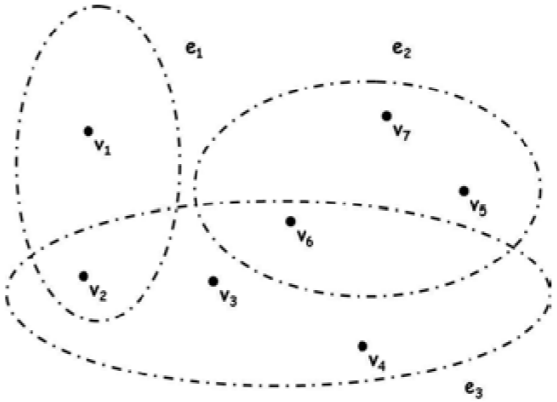


Luo, Haoran, et al. "HyperGraphRAG: Retrieval-Augmented Generation via Hypergraph-Structured Knowledge Representation." *arXiv preprint arXiv:2503.21322* (2025).

Hypergraph Expansions for Message Passing Design

Hypergraph Representation

- **A hypergraph $G = (V, E)$**
  - V : a finite set of objects, E : a family of subsets e of V s.t $\cup_{e \in E} = V$
  - $|V| \times |E|$ matrix H with entries $h(v, e) = 1$ if $v \in e$ and 0 otherwise



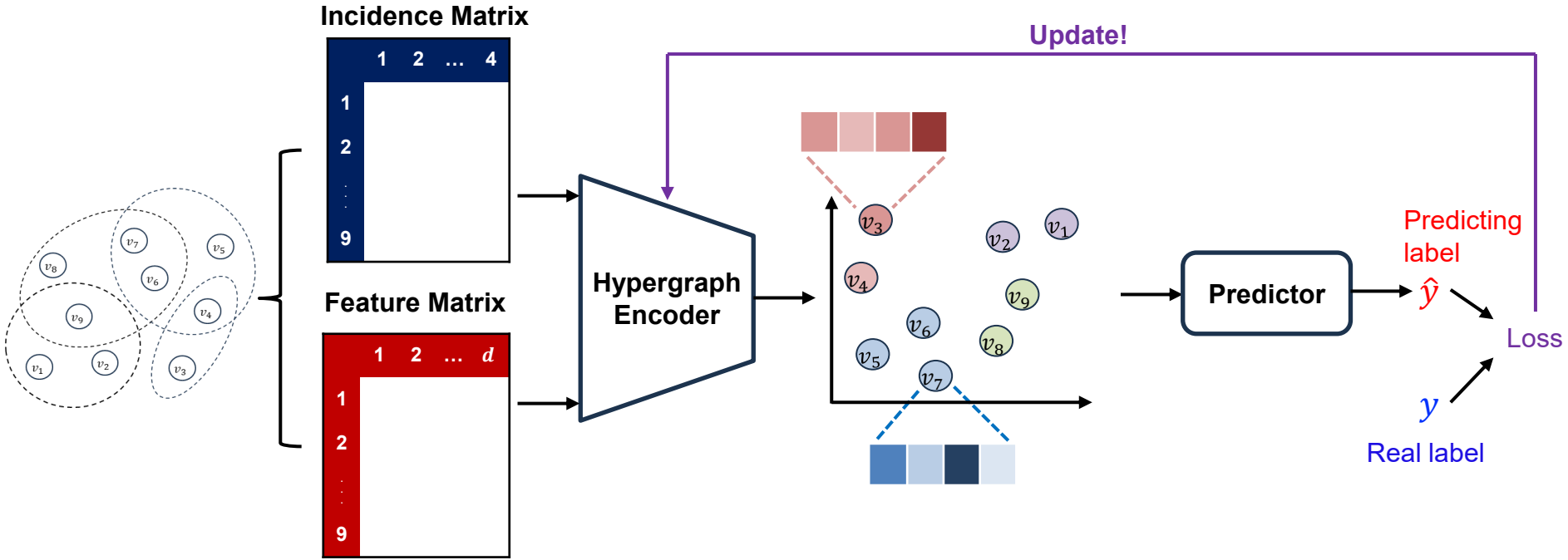
Incidence Matrix

	1	2	3
1	1	0	0
2	1	0	1
3	0	0	1
4	0	0	1
5	0	1	0
6	0	1	1
7	0	1	0

Zhou, Dengyong, Jiayuan Huang, and Bernhard Schölkopf. "Learning with hypergraphs: Clustering, classification, and embedding." *Advances in neural information processing systems* 19 (2006).

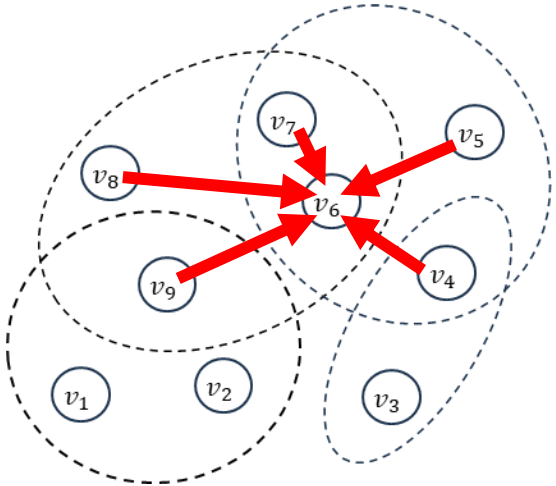
Hypergraph Neural Networks

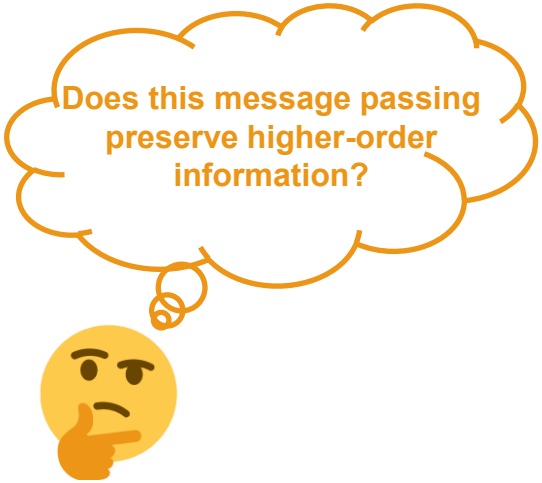
- **Same as graph neural networks**
  - Learn embeddings by **aggregating** and **transforming signals** induced by **incident hyperedges**



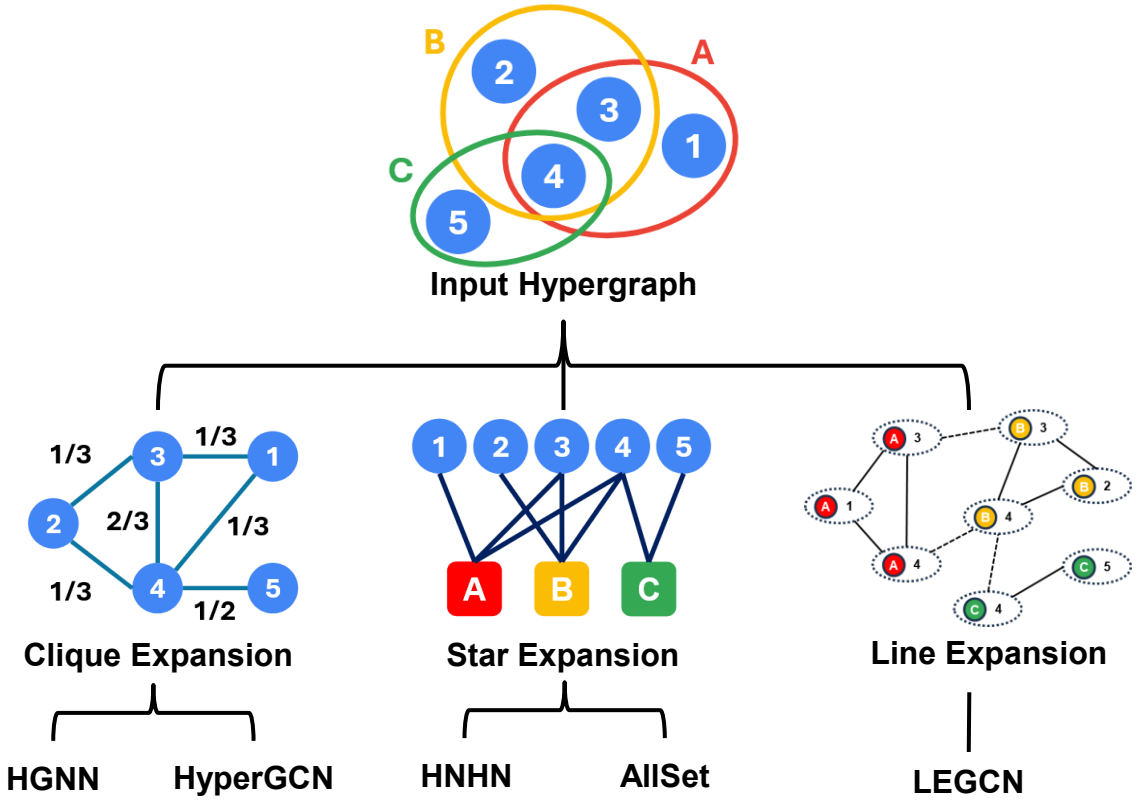
Challenges

- **How to expand hypergraphs into pairwise graphs for message passing?**
  - Preserving higher-order information during the expansion is critical



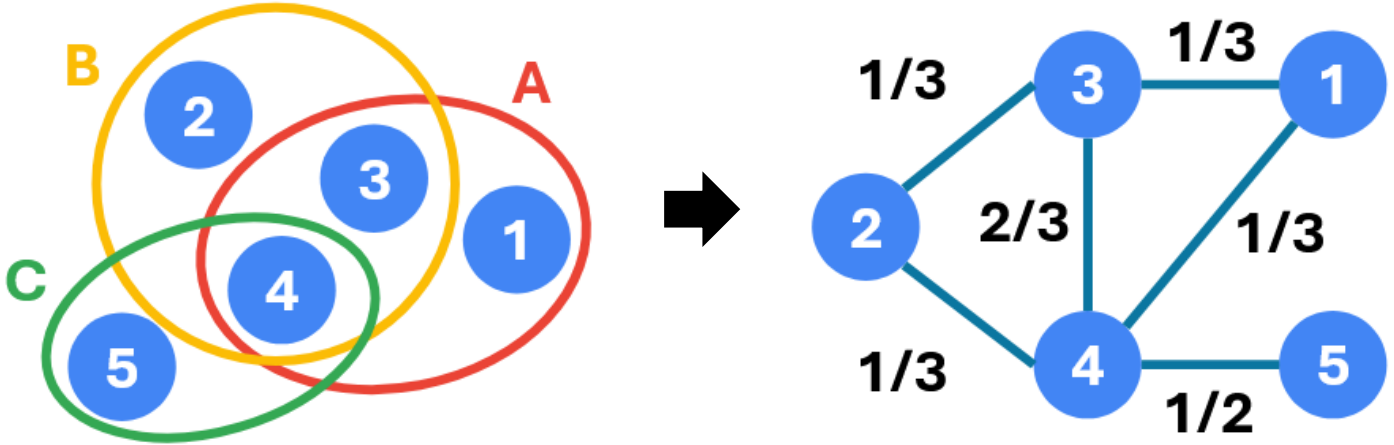


Hypergraph Modeling



Clique Expansion

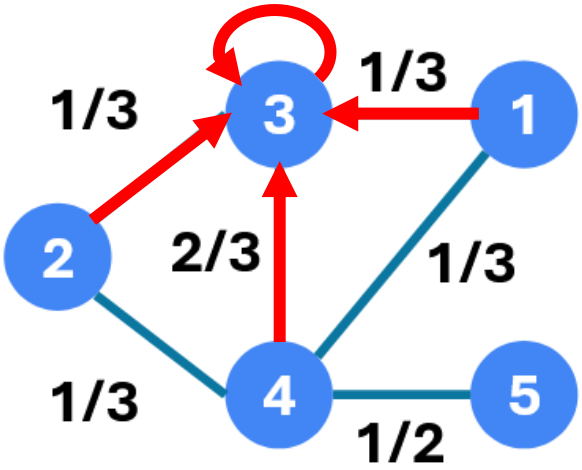
- **Replace each hyperedge with a clique**
  - Node-to-Node
  - Assigning proper edge weights is crucial for capturing HOIs



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

HGNN

- Perform message passing between nodes



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

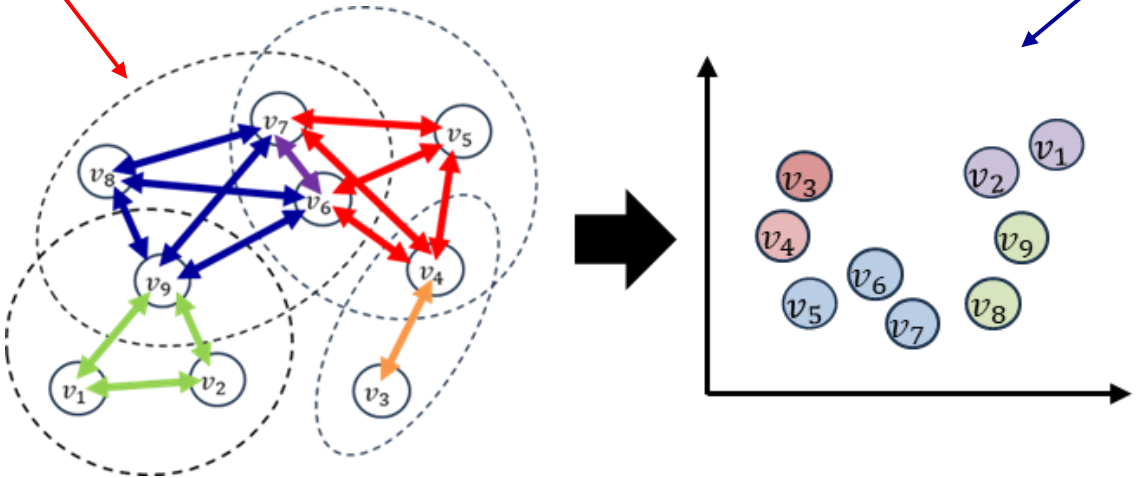
HyperGCN

Motivation

- **Over-smoothing** & **Time complexity** problem can arise

$O(|s|^2)$ per each hyperedge

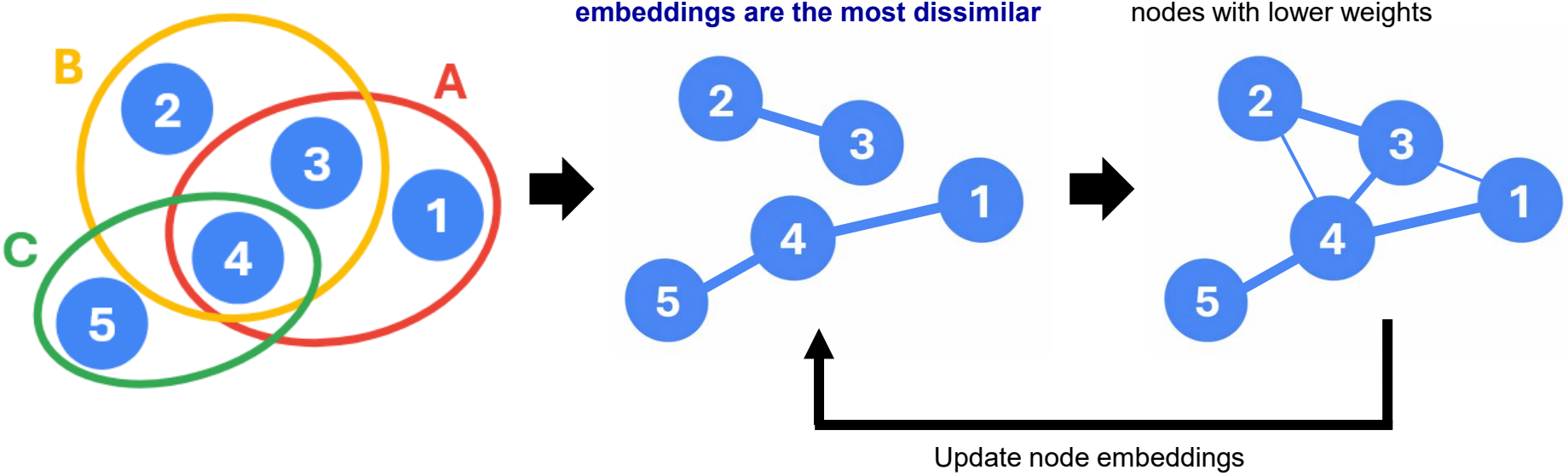
As message passing progresses, all node embeddings become similar



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

HyperGCN

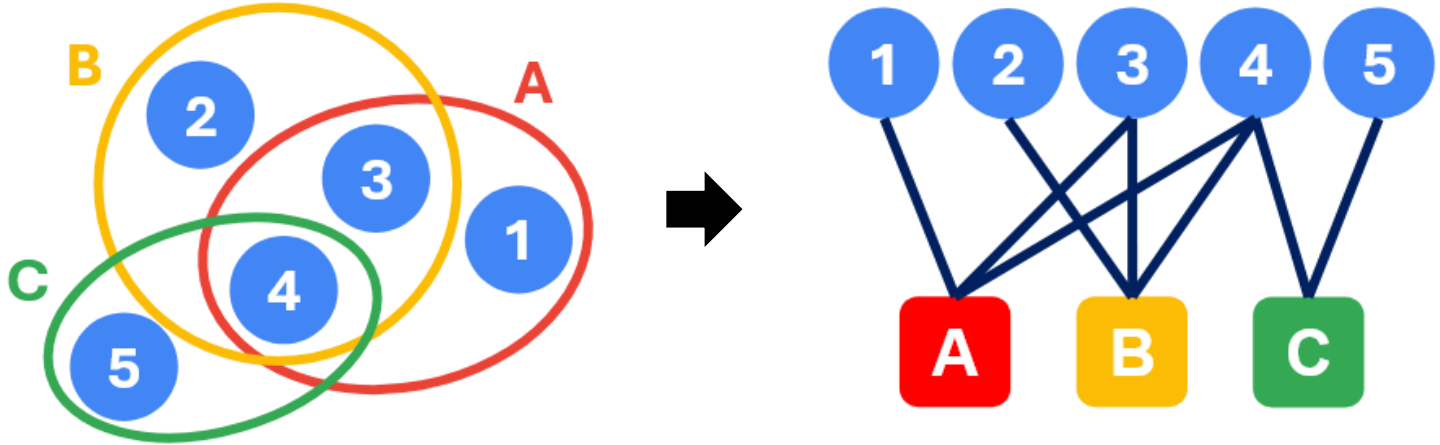
☐ Advanced Clique Expansion



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

Star Expansion

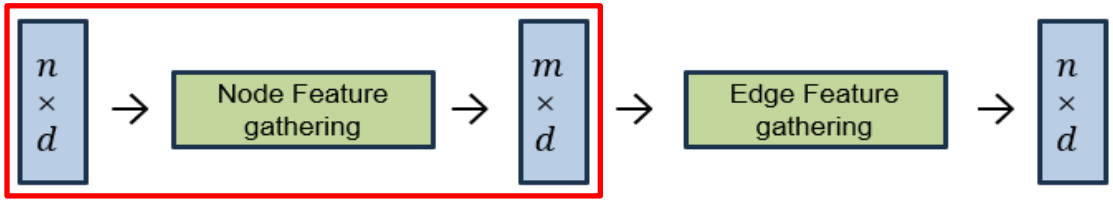
- **View both vertices and hyperedges as nodes**
  - Make **bipartite graph** (vertices ↔ hyperedges)

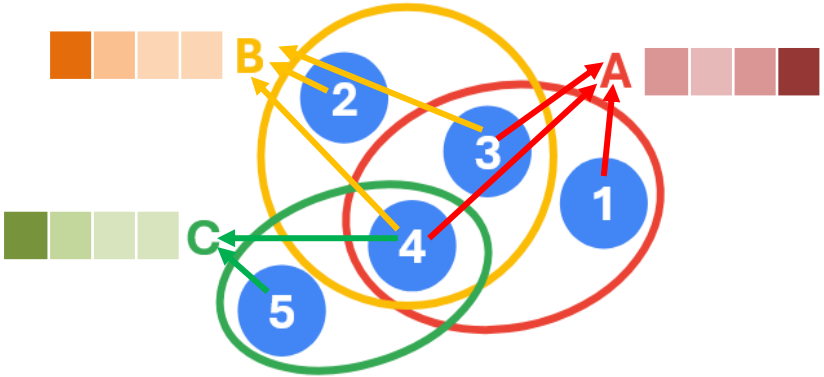


Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

HNHN

- Define hyperedge features through nodes that it contains
  - $X'_E = \sigma(D_{E,l,\beta}^{-1} A D_{V,r,\beta} X_V W_E + b_E)$

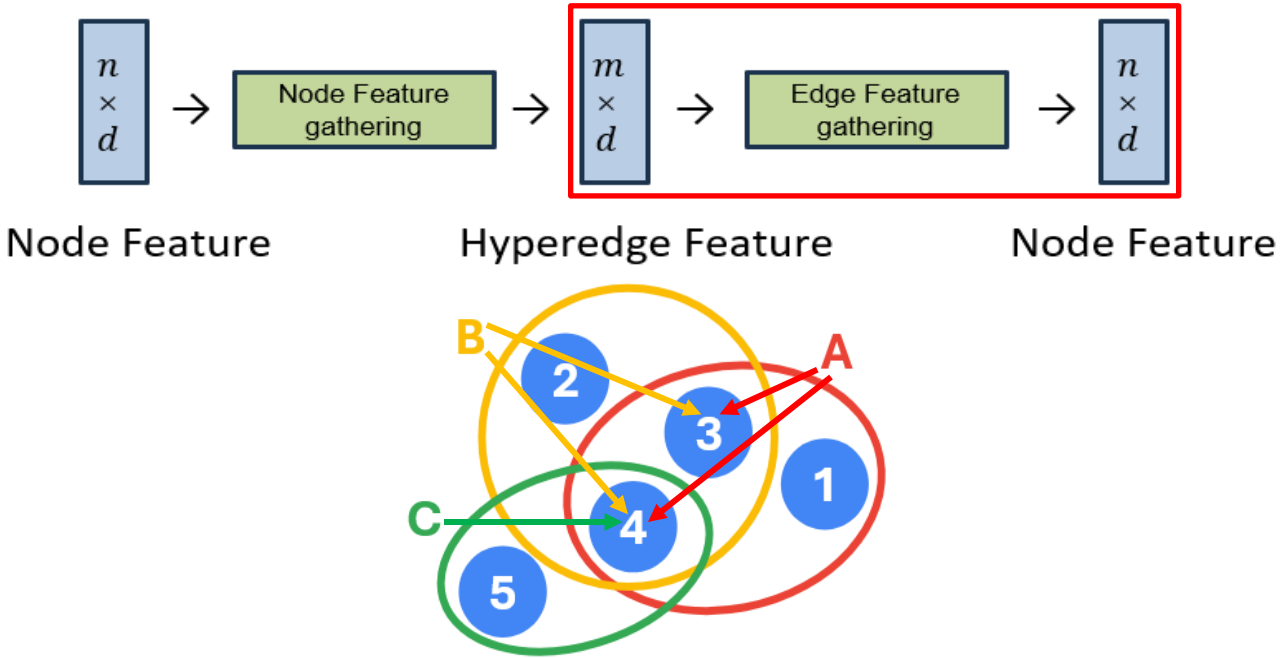




Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

HNHN

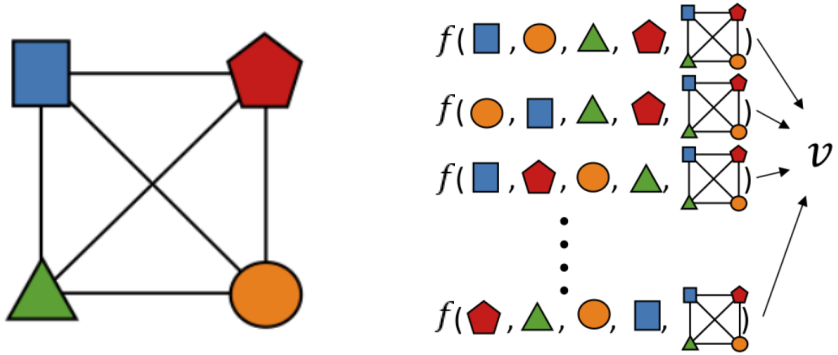
- Define nodes through hyperedges that contain itself
  - $X'_V = \sigma(D_{V,l,\alpha}^{-1} A D_{E,r,\alpha} X'_E W_V + b_V)$



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

AllSet

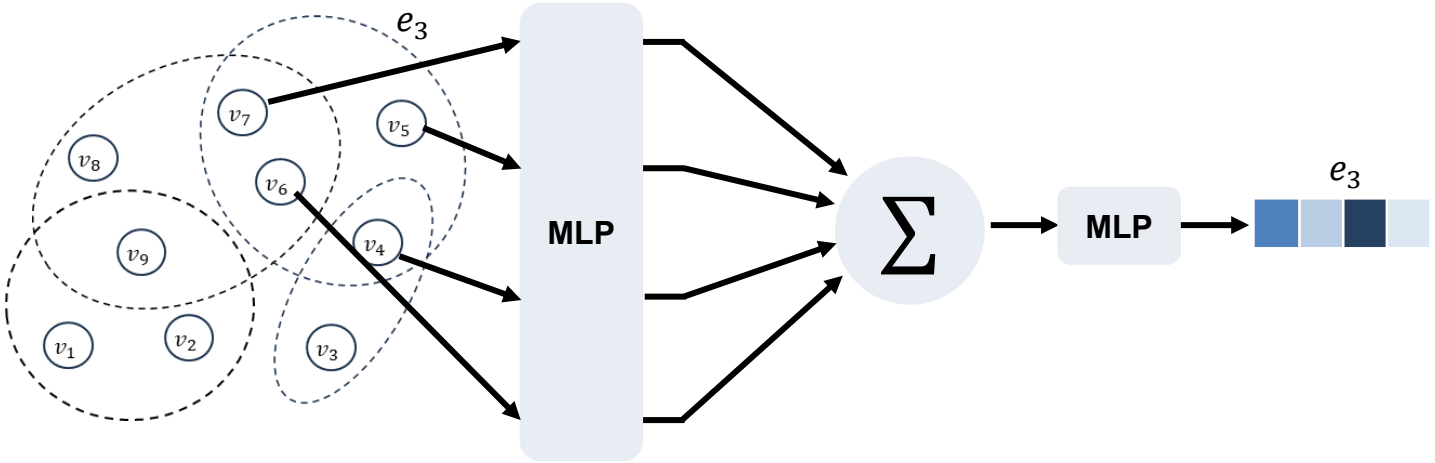
- **What is a multiset function?**
  - *A function f is a multiset function if it is permutation invariant*
  - **A multiset function can formulate aggregation for message passing**



AllSet

Key Idea

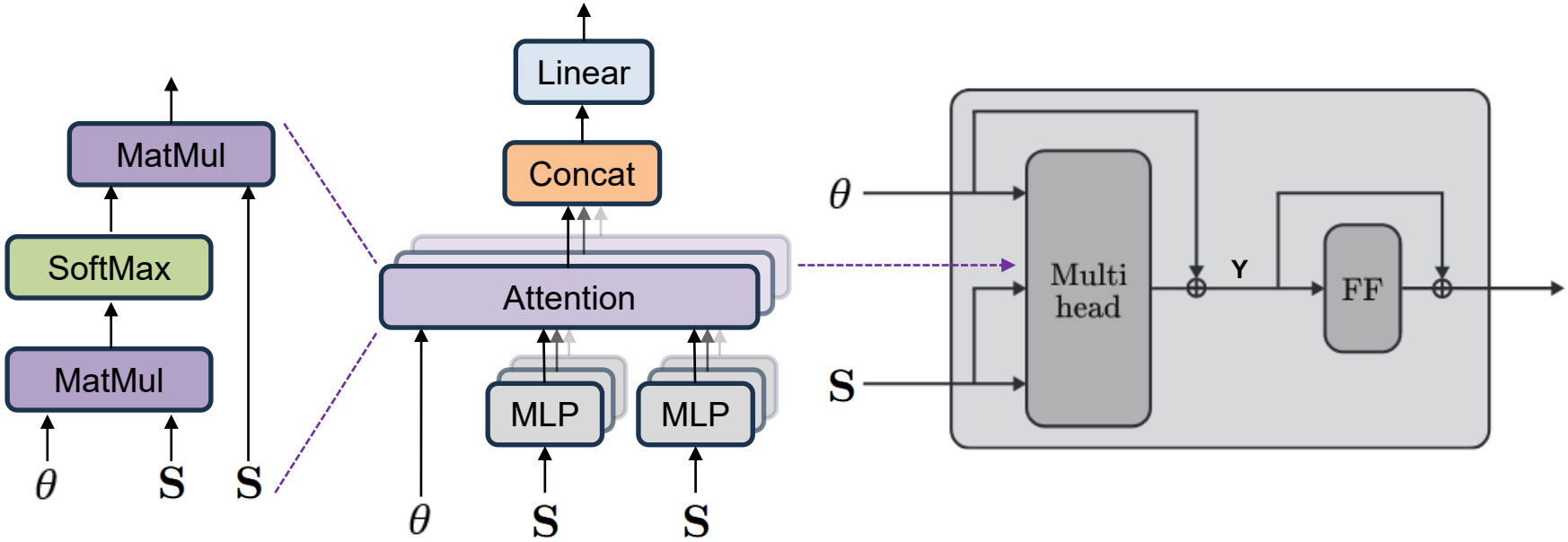
- Any multiset functions f can be parametrized
  - $f(S) = \rho(\sum_{s \in S} \phi(s))$, where ρ and ϕ are some bijective mappings
  - These mappings can be replaced by any universal approximator such as a MLP



AllSet

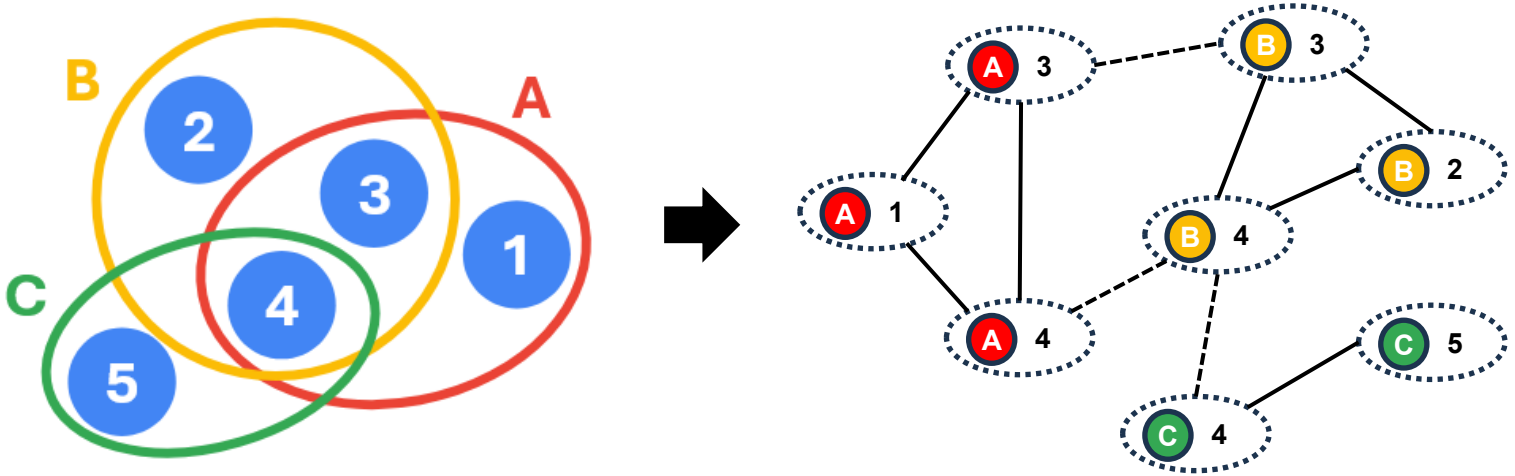
AllSetTransformer

- Attention-based AllSet layer for hypergraph neural networks



Line Expansion

- **View node-hyperedge relations as nodes**
  - View each node as **a vertex with hyperedge context** or **a hyperedge with vertex context**



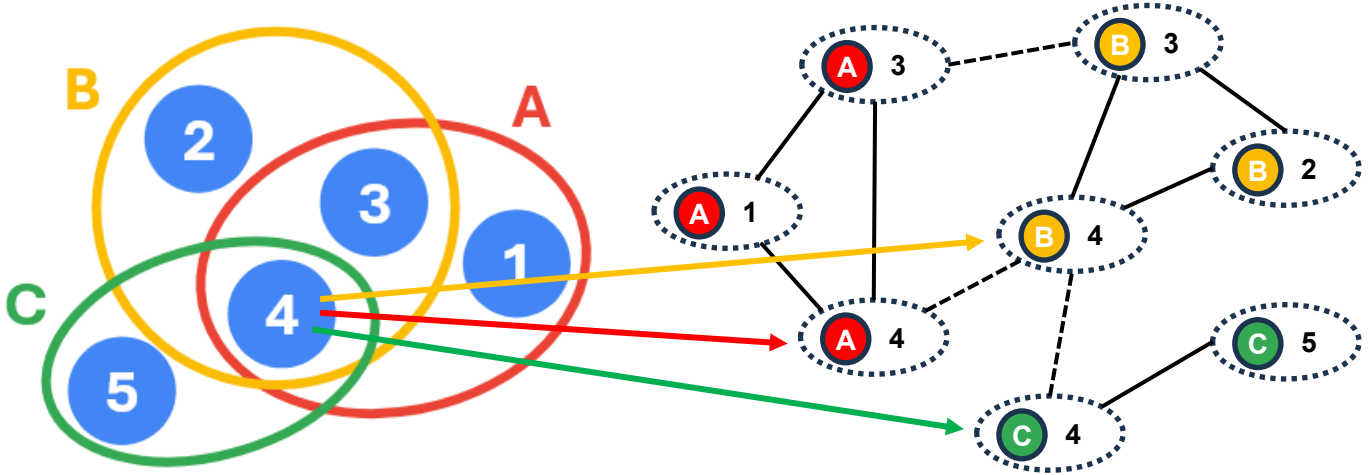
Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

LEGCN

Feature Projection

- Scatter features from vertex of $\mathcal{G}_H$ to feature vectors of line nodes in $\mathcal{G}_l$

$$H^{(0)} = P_{vertex}X \in \mathbb{R}^{|\mathcal{V}_l| \times d_i}$$



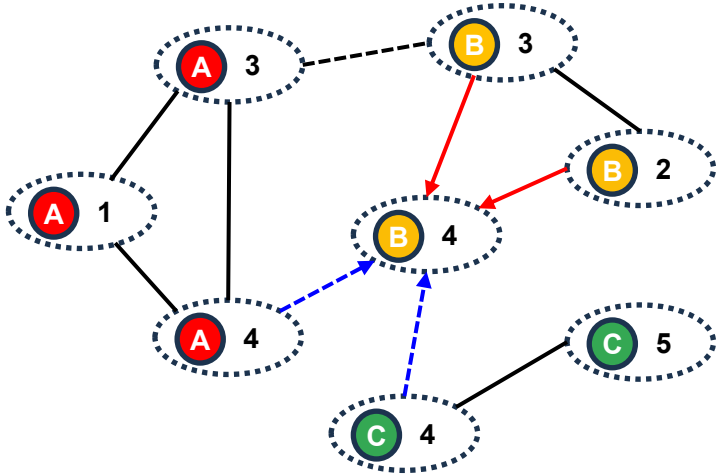
Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

LEGCN

Feature aggregation

- Convolve information from neighbors who share the same hyperedges(red)/vertices(blue)

$$h_{(v,e)}^{(k+1)} = \sigma\Big(\sum_{e'} w_e h_{(v,e')}^{(k)} \Theta^{(k)} + \sum_{v'} w_v h_{(v',e)}^{(k)} \Theta^{(k)}\Big)$$

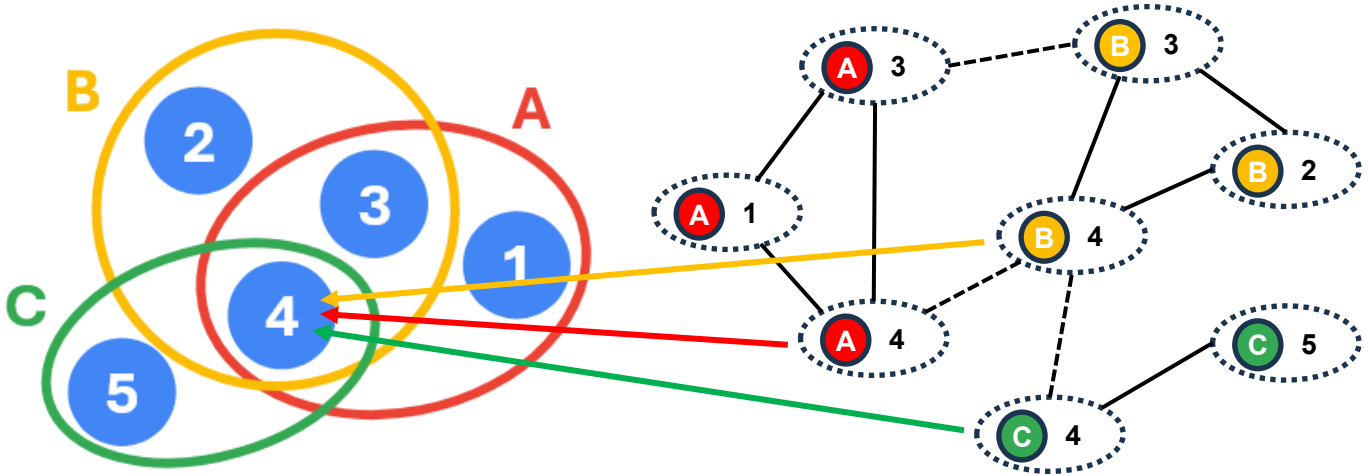


LEGCN

Representation Back-projection

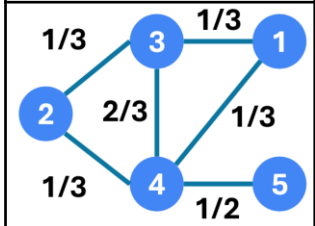
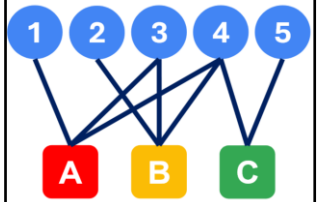
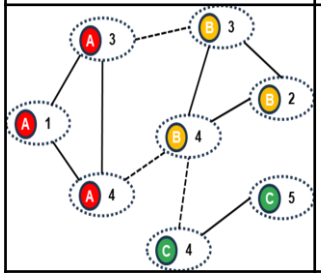
- Fuse the learned representation using P'_{vertex} in an inverse edge degree manner

$$Y = P'_{vertex} H^{(K)} \in \mathbb{R}^{|\mathcal{V}| \times d_o}$$



Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

Comparison of Expansion Schemes

Expansion	Graph Construction	Message passing	Pros	Cons
	Each hyperedge $e \rightarrow$ clique	**Node-to-Node**	Various GNN strategies can be directly applied	Higher-order information can be lost
	Add a **hyperedge node** for each hyperedge	**Node-to-Hyperedge**	Can maintain higher-order information	Direct GNN techniques can be non-trivial
	Each incidence (v, e) becomes a **line node**	**Incidence-to-Incidence**	Can maintain higher-order information	Memory Issue

Kim, Sunwoo, et al. "A survey on hypergraph neural networks: an in-depth and step-by-step guide." *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 2024.

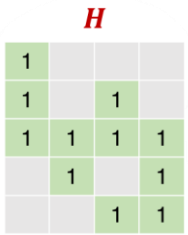
Self-Supervised Hypergraph Learning

Problems

- Hypergraph data labeling is often time, resource, and labor-intensive

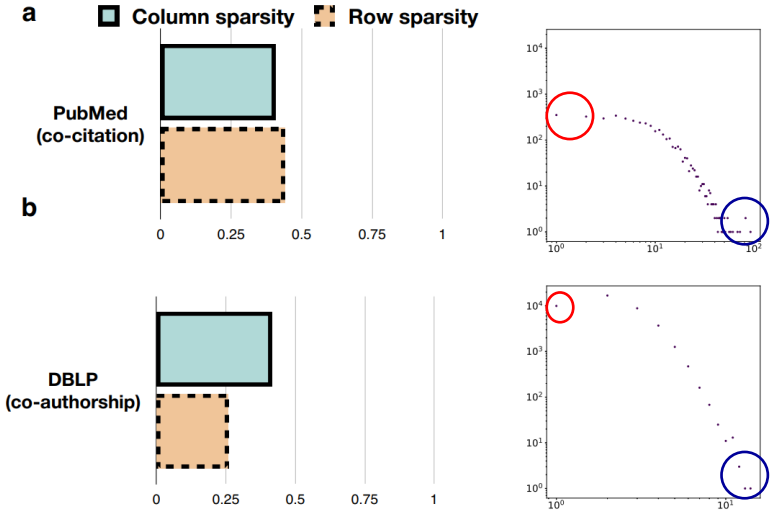
Problems

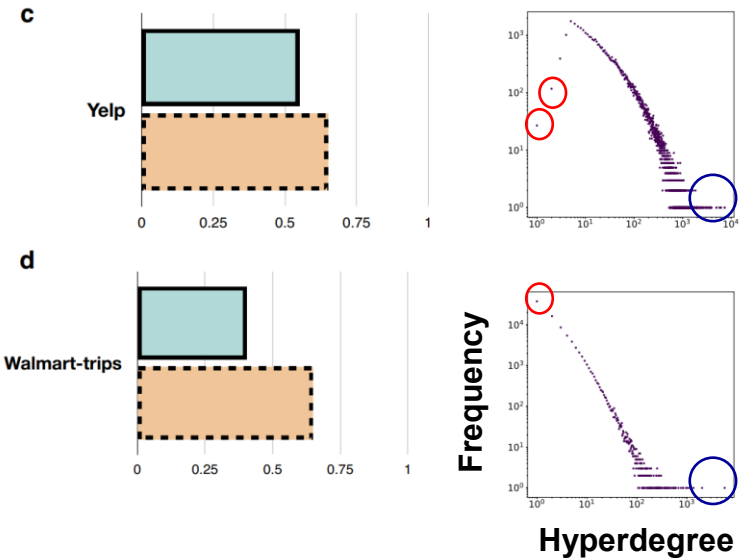
- **Real world (hyper)graphs are sparse**
  - Most objects have only a few relationships



: Low-degree nodes

: High-degree nodes



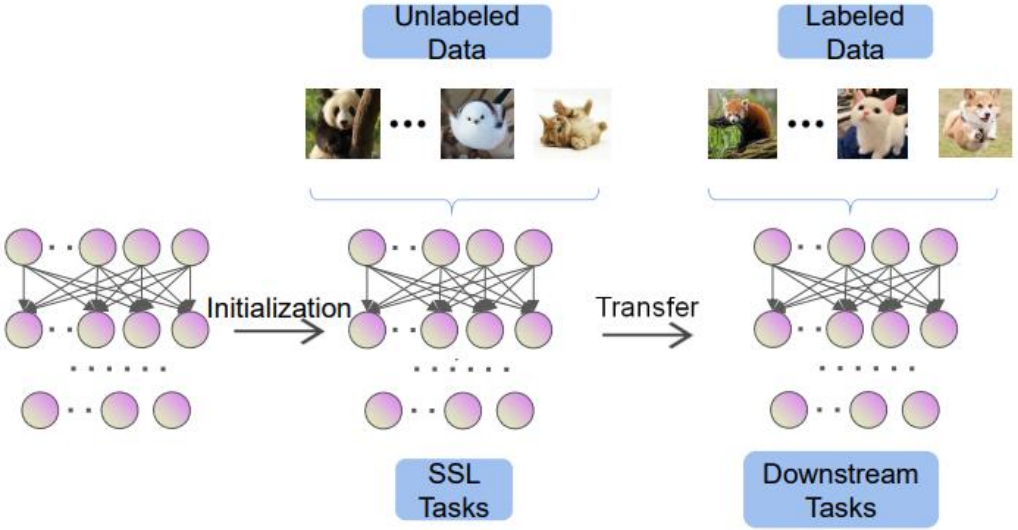


Ye, Bodian, et al. "EasyHypergraph: an open-source software for fast and memory-saving analysis and learning of higher-order networks." *Humanities and Social Sciences Communications* 12.1 (2025): 1-19.

Solution

Self-Supervised Learning (SSL)

- Learn discriminative features from vast quantities of unlabeled instances without relying on human annotations

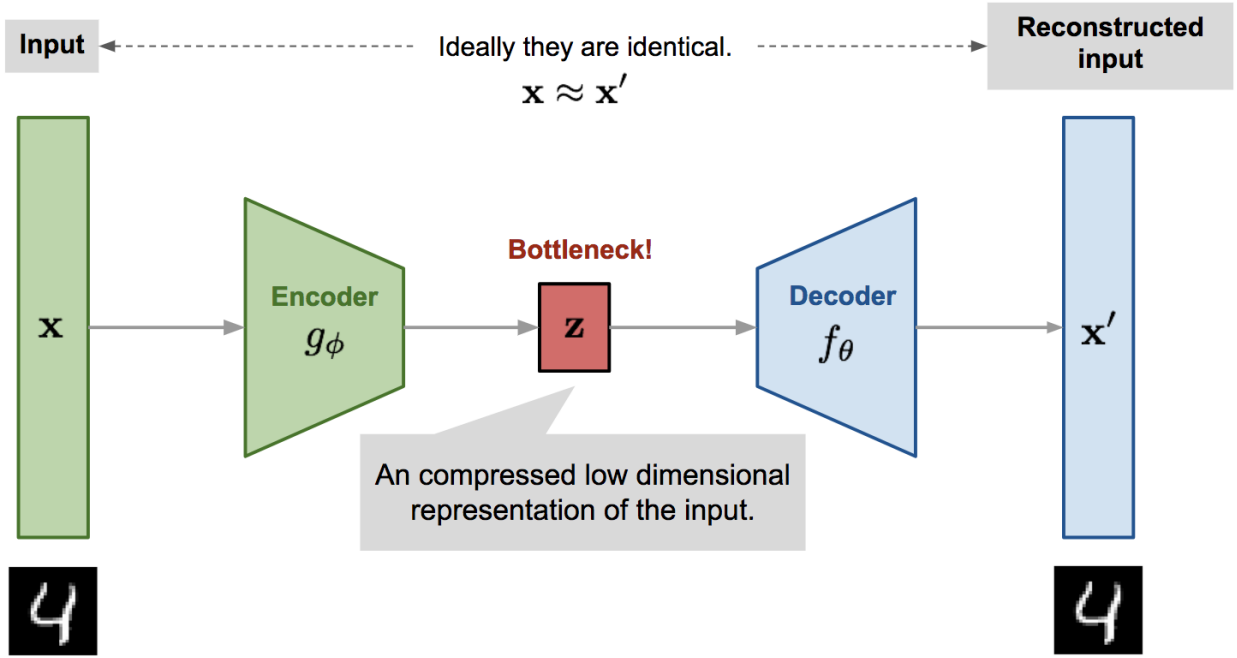


Gui, Jie, et al. "A survey on self-supervised learning: Algorithms, applications, and future trends." *IEEE Transactions on Pattern Analysis and Machine Intelligence* 46.12 (2024): 9052-9071.

Generation-based Methods

Key Idea

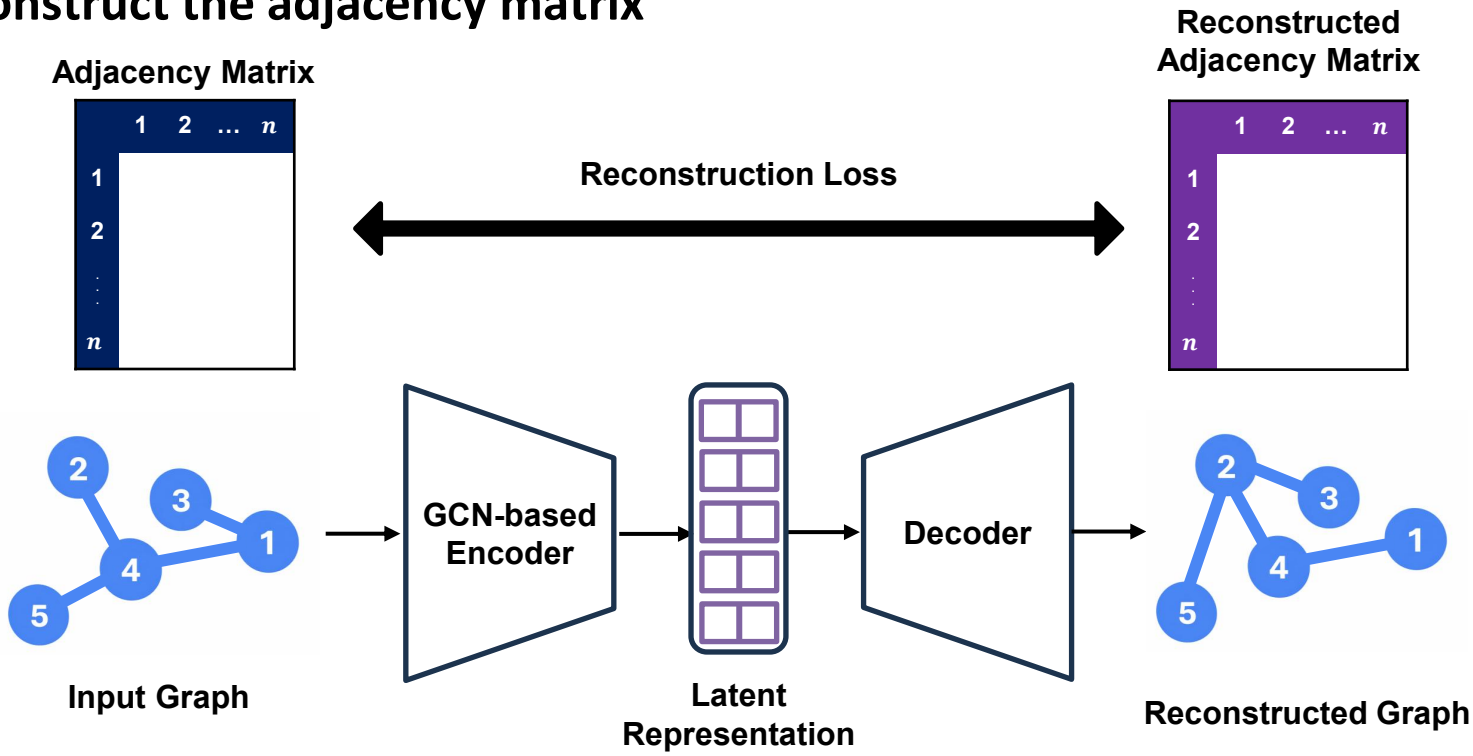
- Aim to **reconstruct the input data**



https://lilianweng.github.io/posts/2018-08-12-vae/

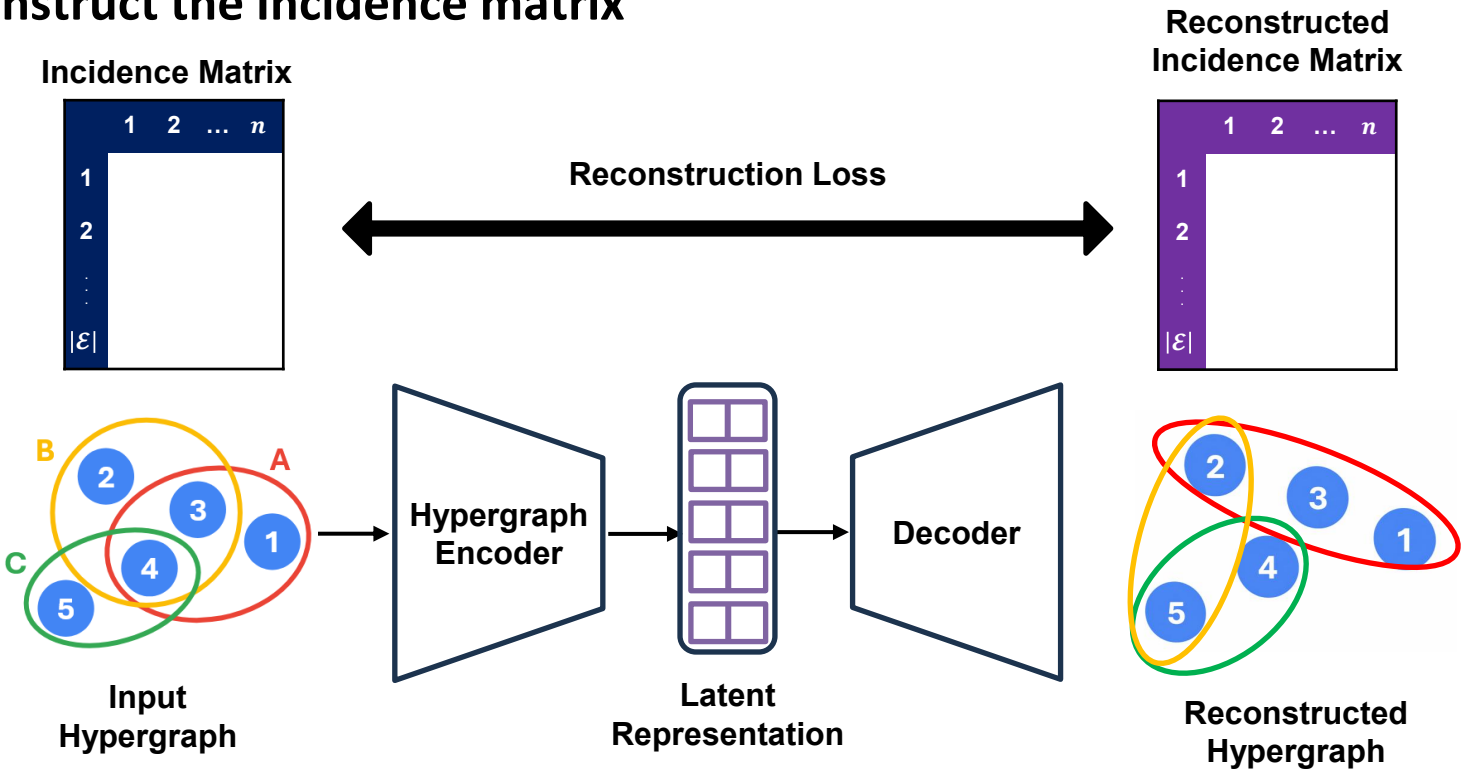
Generation-based Graph SSL

- Reconstruct the adjacency matrix



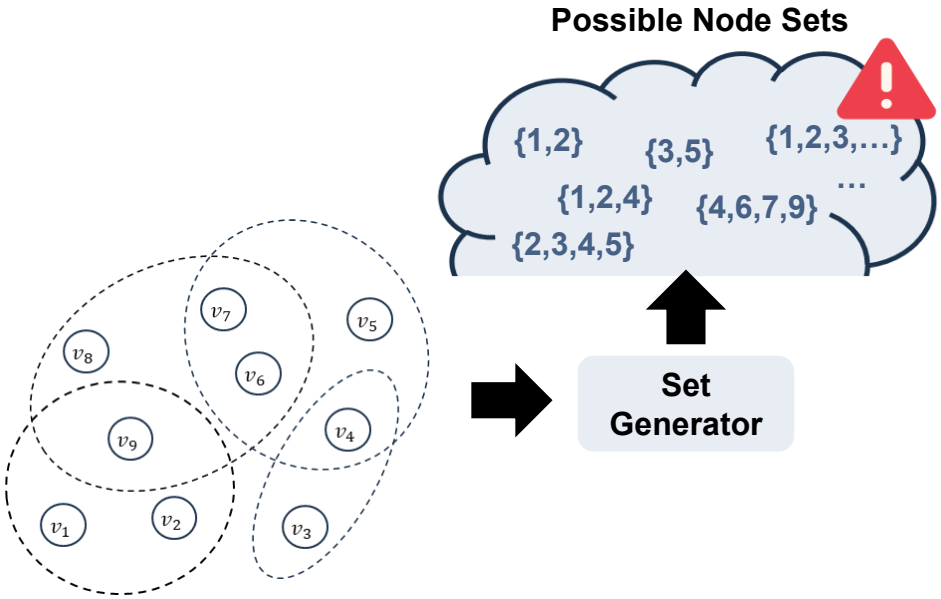
Generation-based Hypergraph SSL

- Reconstruct the Incidence matrix



Challenges

- **How can we reconstruct incidence?**
  - Full incidence-matrix recovery is combinatorial and intractable



HypeBoy: Generative Self-Supervised Representation Learning on Hypergraphs

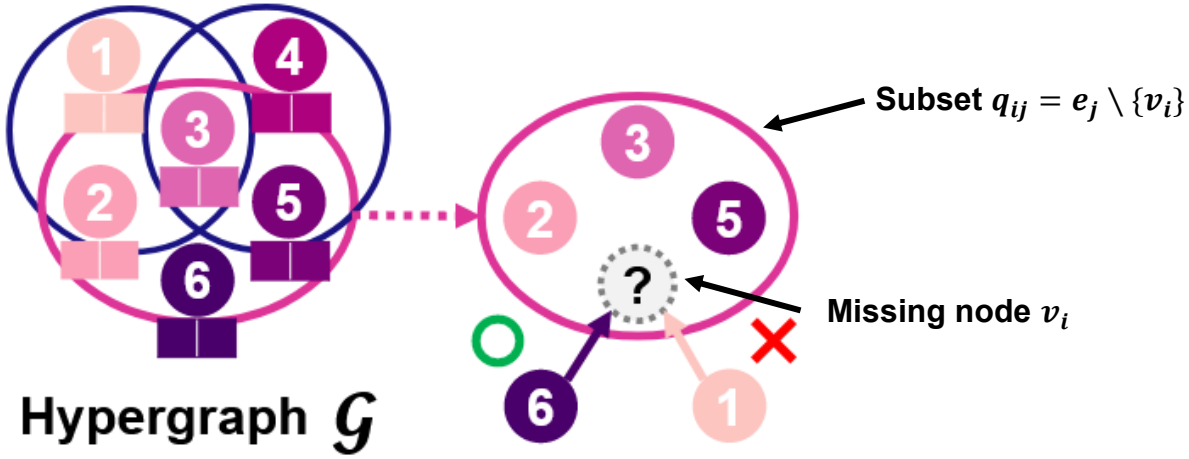
Sunwoo Kim, Shinhwan Kang, Fanchen Bu, Soo Yong Lee, Jaemin Yoo, Kijung Shin

Korea Advanced Institute of Science and Technology (KAIST)

24-ICLR

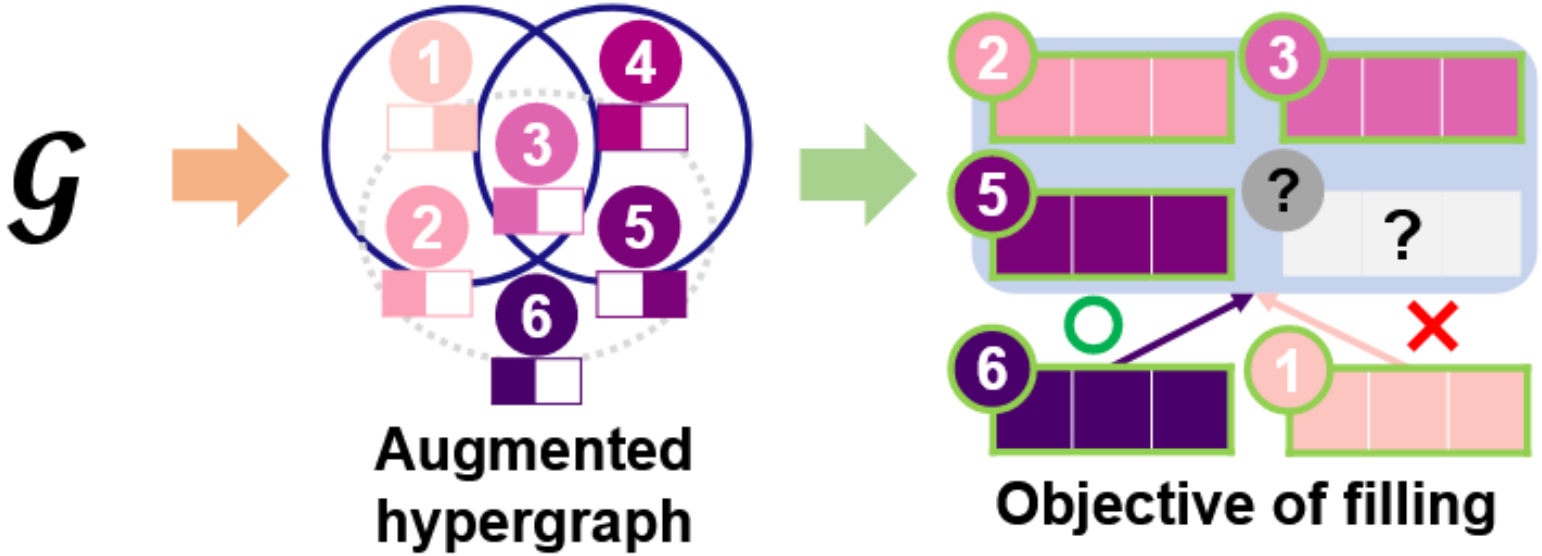
Hyperedge Filling

- **Aim to predict a node that is likely to from a hyperedge with it**
  - Maximize the probability of v_i correctly completing q_{ij}



Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Hypeboy: Hypergraphs, build own hyperedges

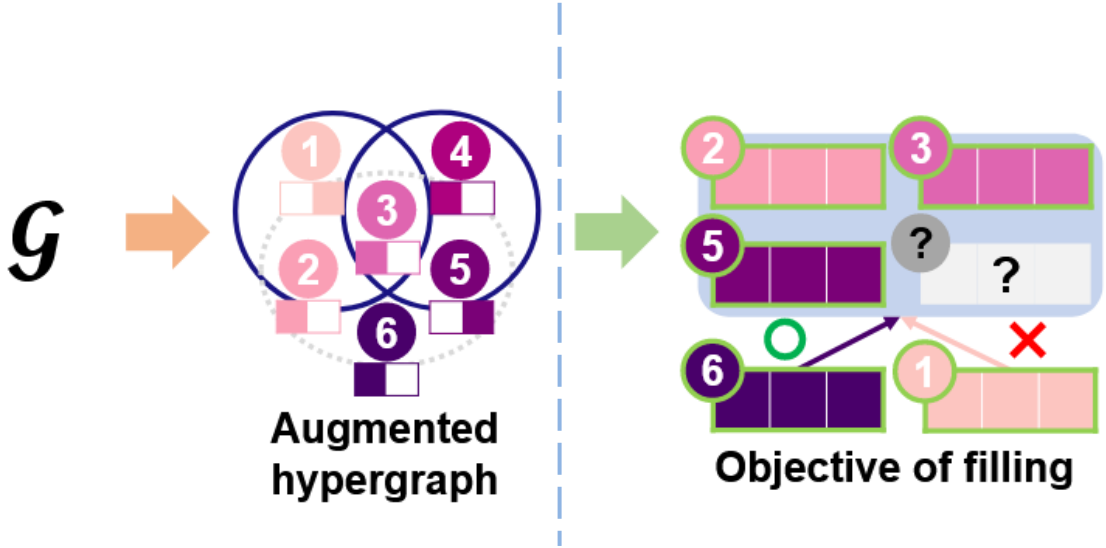


Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Hypeboy: Hypergraphs, build own hyperedges

Hypergraph Augmentation

- Compose of feature masking and hyperedge sampling
- Allow the encoder to not overfit to the neighbor distribution and features

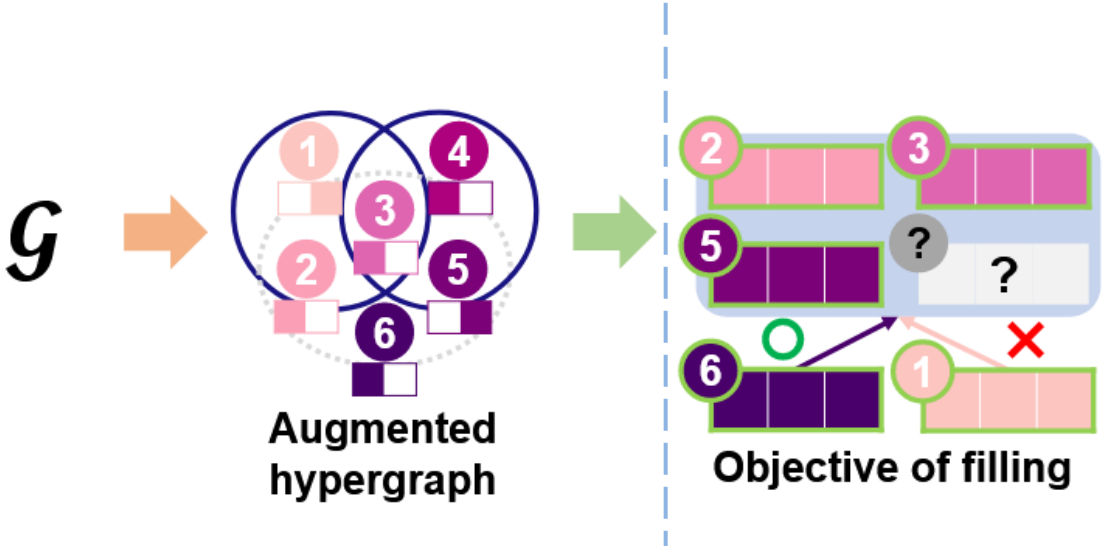


Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Hypeboy: Hypergraphs, build own hyperedges

Hypergraph Encoding

- Encode nodes and queries with an HNN encoder and projection heads
- Prevent dimensional collapse in node representations

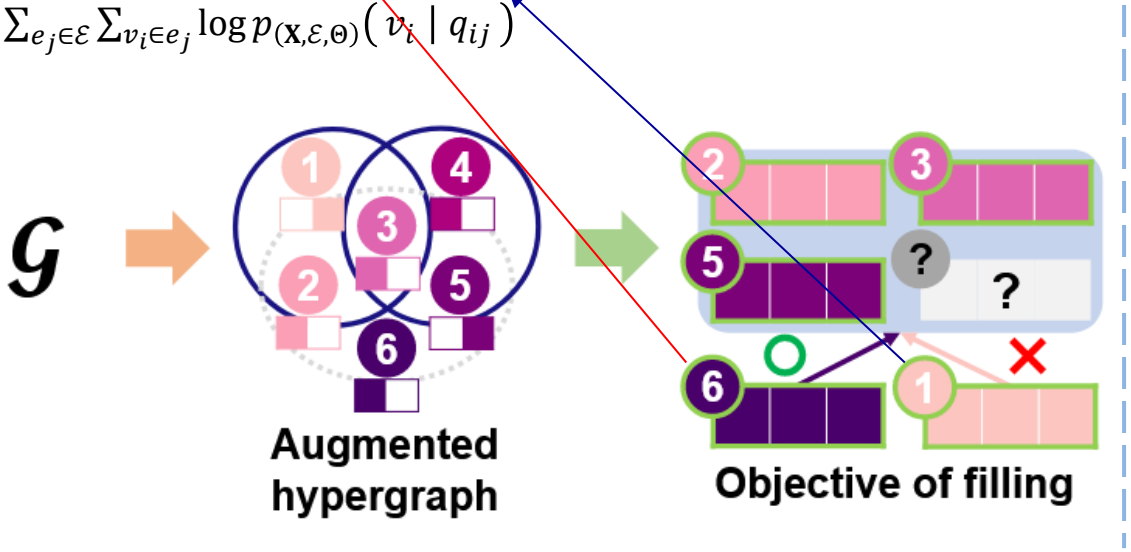


Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Hypeboy: Hypergraphs, build own hyperedges

Hyperedge Filling Loss

- Compute the SSL loss based on the hyperedge filling probability
  - $p_{(\mathbf{X},\mathcal{E},\Theta)}(v_i \mid q_{ij}) := \frac{\exp(sim(\boldsymbol{h_i},\boldsymbol{q_{ij}}))}{\sum_{v_k \in \mathcal{V}} \exp(sim(\boldsymbol{h_k},\boldsymbol{q_{ij}}))}$
  - $\mathcal{L} := -\sum_{e_j \in \mathcal{E}} \sum_{v_i \in e_j} \log p_{(\mathbf{X},\mathcal{E},\Theta)}(v_i \mid q_{ij})$



Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Hypeboy: Hypergraphs, build own hyperedges

Performance on Node Classification

- Show the best average ranking among all 17 methods

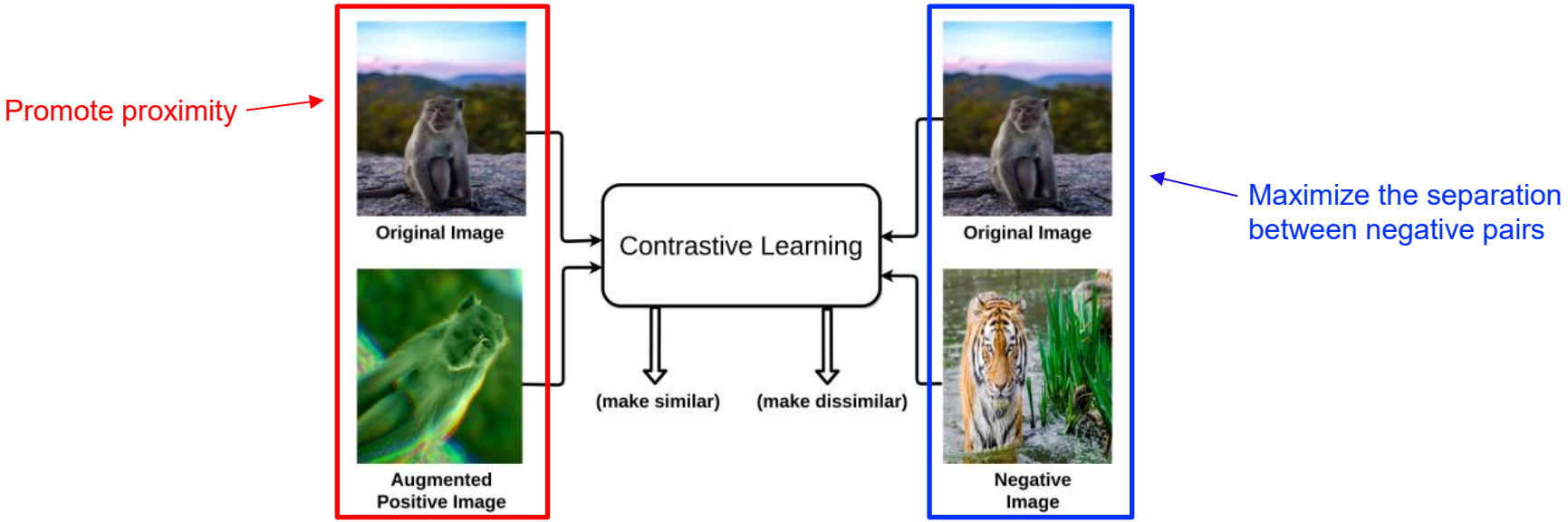
	Method	Citeseer	Cora	Pubmed	Cora-CA	DBLP-P	DBLP-A	AMiner	IMDB	MN-40	20News	House	A.R.
(Semi-)Supervised	R.G.	18.1 (0.9)	17.4 (1.0)	36.0 (0.7)	17.8 (0.7)	18.9 (0.2)	25.6 (1.0)	10.2 (0.2)	33.9 (0.7)	3.7 (0.2)	50.1 (1.3)	26.7 (0.3)	16.9
	MLP	32.5 (7.0)	27.9 (7.0)	62.1 (3.7)	34.8 (5.1)	73.5 (1.0)	56.0 (4.9)	22.3 (1.7)	39.1 (2.4)	89.4 (1.5)	73.1 (1.4)	72.2 (3.9)	13.4
	HGNN	41.9 (7.8)	50.0 (7.2)	72.9 (5.0)	50.2 (5.7)	85.3 (0.8)	67.1 (6.0)	30.3 (2.5)	42.2 (2.9)	88.0 (1.4)	76.4 (1.9)	52.7 (3.8)	10.0
	HyperGCN	31.4 (9.5)	33.1 (10.2)	63.5 (14.4)	37.1 (9.1)	53.5 (11.6)	68.2 (14.4)	26.4 (3.6)	37.9 (4.5)	55.1 (7.8)	67.0 (9.4)	49.8 (3.5)	14.7
	HNHN	43.1 (8.7)	50.0 (7.9)	72.1 (5.4)	48.3 (6.2)	84.6 (0.9)	62.6 (4.8)	30.0 (2.4)	42.3 (3.4)	86.1 (1.6)	74.2 (1.5)	49.7 (2.2)	11.9
	UniGCN	44.2 (8.1)	49.1 (8.4)	74.4 (3.9)	51.3 (6.3)	86.9 (0.6)	65.1 (4.7)	32.7 (1.8)	41.6 (3.5)	89.1 (1.0)	77.2 (1.2)	51.1 (2.4)	9.1
	UniGIN	40.4 (9.1)	47.8 (7.7)	69.8 (5.6)	48.3 (6.1)	83.4 (0.8)	63.4 (5.1)	30.2 (1.4)	41.4 (2.7)	88.2 (1.8)	70.6 (1.8)	51.1 (3.0)	13.0
	UniGCNII	44.2 (9.0)	48.5 (7.4)	74.1 (3.9)	54.8 (7.5)	87.4 (0.6)	65.8 (3.9)	32.5 (1.7)	42.5 (3.9)	90.8 (1.1)	70.9 (1.0)	50.8 (4.3)	8.8
	AllSet	43.5 (8.0)	47.6 (4.2)	72.4 (4.5)	57.5 (5.7)	85.9 (0.6)	65.3 (3.9)	29.3 (1.2)	42.3 (2.4)	92.1 (0.6)	71.9 (2.5)	54.1 (3.4)	9.8
	ED-HNN	40.3 (8.0)	47.6 (7.7)	72.7 (4.7)	54.8 (5.4)	86.2 (0.8)	65.8 (4.8)	30.0 (2.1)	41.4 (3.0)	90.7 (0.9)	76.2 (1.2)	71.3 (3.7)	9.6
	PhenomNN	49.8 (9.6)	56.4 (9.6)	76.1 (3.5)	60.8 (6.2)	88.1 (0.4)	72.3 (4.1)	33.8 (2.0)	44.1 (3.7)	95.9 (0.8)	74.0 (1.5)	70.4 (5.6)	3.9
Self-supervised	GraphMAE2	41.1 (10.0)	49.3 (8.3)	72.9 (4.2)	55.4 (8.4)	86.6 (0.6)	69.5 (4.4)	32.8 (1.9)	43.3 (2.7)	90.1 (0.7)	71.9 (1.3)	52.8 (3.5)	8.4
	MaskGAE	49.6 (10.1)	57.1 (8.8)	72.8 (4.3)	57.8 (5.9)	86.3 (0.5)	74.8 (3.1)	33.7 (1.6)	44.5 (2.5)	90.0 (0.9)	O.O.T.	51.8 (3.3)	7.4
	TriCL	51.7 (9.8)	60.2 (7.9)	76.2 (3.6)	64.3 (5.5)	88.0 (0.4)	79.7 (2.9)	33.1 (2.2)	46.9 (2.9)	90.3 (1.0)	77.2 (1.0)	69.7 (4.9)	3.4
	HyperGCL	47.0 (9.2)	60.3 (7.4)	76.8 (3.7)	62.0 (5.1)	87.6 (0.5)	79.7 (3.8)	33.2 (1.6)	43.9 (3.6)	91.2 (0.8)	77.8 (0.8)	69.2 (4.9)	3.5
	H-GD	45.4 (9.9)	50.6 (8.2)	74.5 (3.5)	58.8 (6.2)	87.3 (0.5)	75.1 (3.6)	32.6 (2.2)	43.0 (3.3)	90.0 (1.0)	77.2 (1.0)	69.7 (5.1)	5.8
	HYPEBOY	56.7 (9.8)	62.3 (7.7)	77.0 (3.4)	66.3 (4.6)	88.2 (0.4)	80.6 (2.3)	34.1 (2.2)	47.6 (2.5)	90.4 (0.9)	77.6 (0.9)	70.4 (4.8)	1.7

Kim, Sunwoo, et al. "Hypeboy: Generative self-supervised representation learning on hypergraphs." *arXiv preprint arXiv:2404.00638* (2024).

Contrast-based Methods

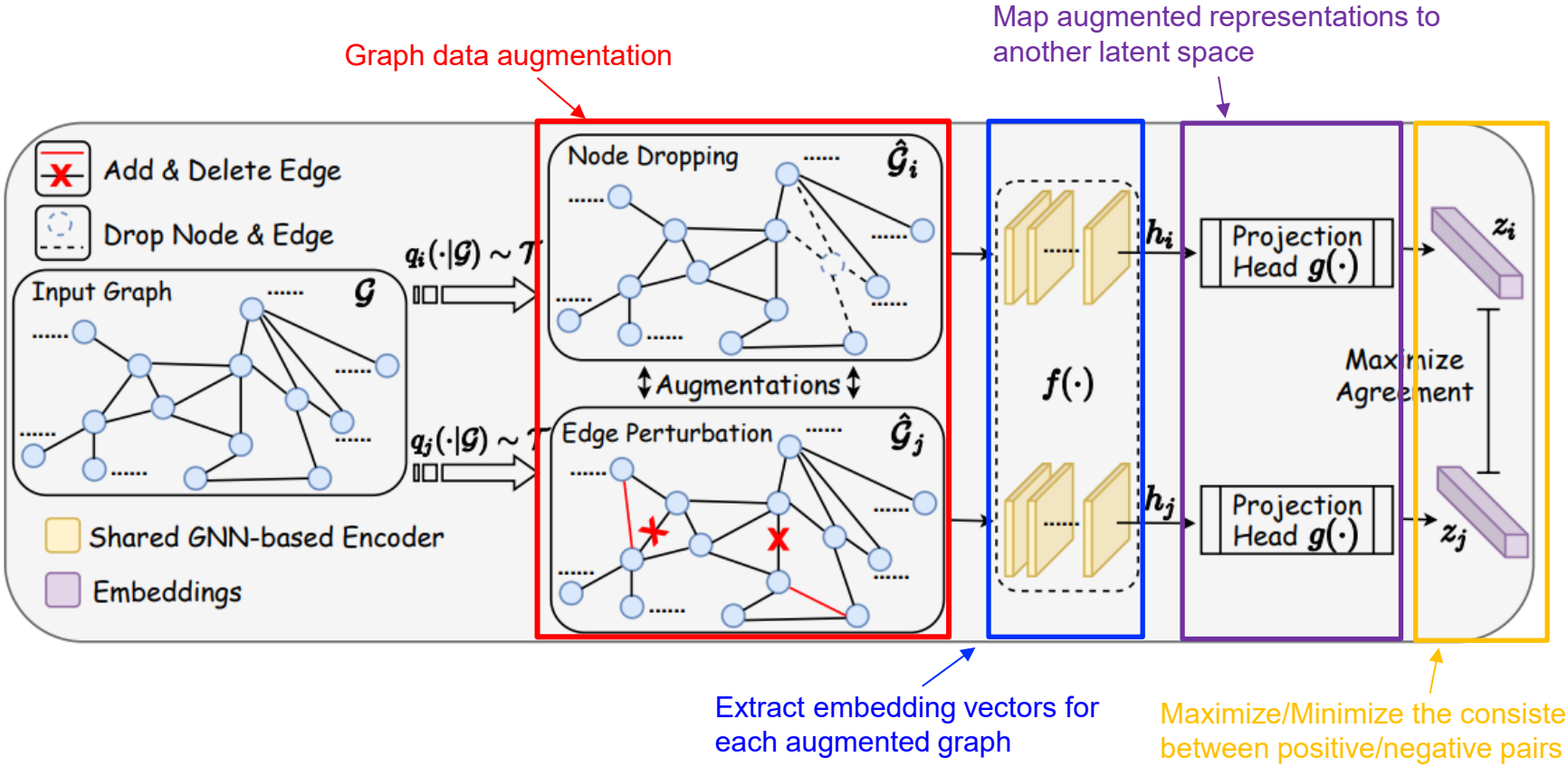
Key Idea

- **Push original and augmented data closer, push original and negative data away**



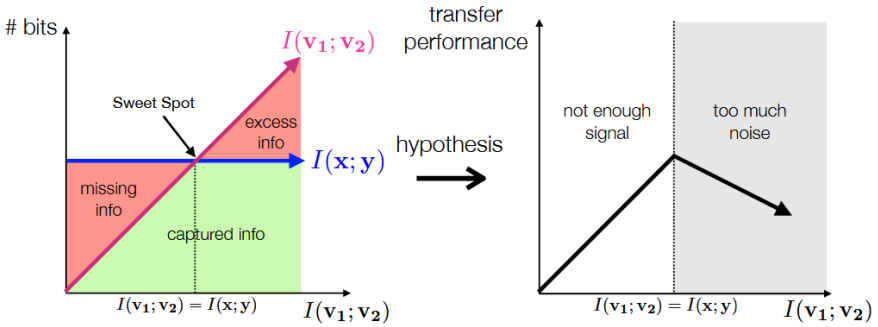
Jaiswal, Ashish, et al. "A survey on contrastive self-supervised learning." *Technologies* 9.1 (2020): 2.

Contrast-based Graph SSL

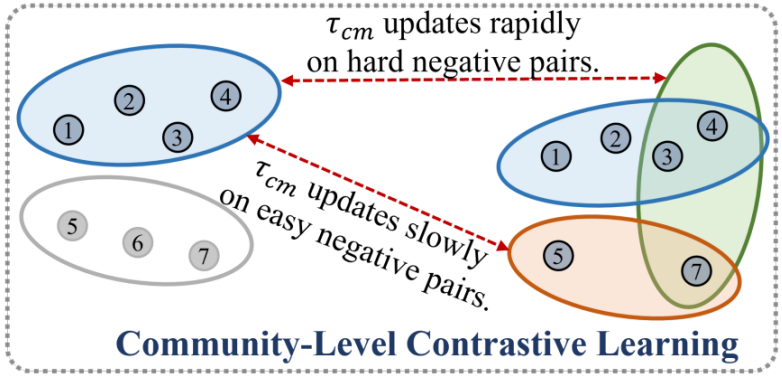


Challenges

- **How to augment a hypergraph?**
  - The choice of views is what controls the information the representation captures



- **What to contrast?**
  - Node-only contrast cannot reflect higher-order information



Tian, Yonglong, et al. "What makes for good views for contrastive learning?." *Advances in neural information processing systems* 33 (2020): 6827-6839.
Qian, Yiyue, et al. "Dual-level hypergraph contrastive learning with adaptive temperature enhancement." *Companion Proceedings of the ACM Web Conference 2024*. 2024.

I’m Me, We’re Us, and I’m Us: Tri-directional Contrastive Learning on Hypergraphs

Dongjin Lee and Kijung Shin

School of Electrical Engineering, Kim Jaechul Graduate School of AI, KAIST, South Korea

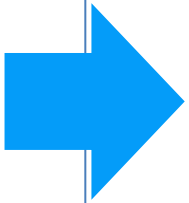
23-AAAI

TriCL

☐ Motivation

Node-only Contrast

- Cannot capture higher-order relations
- Lead to limited expressiveness



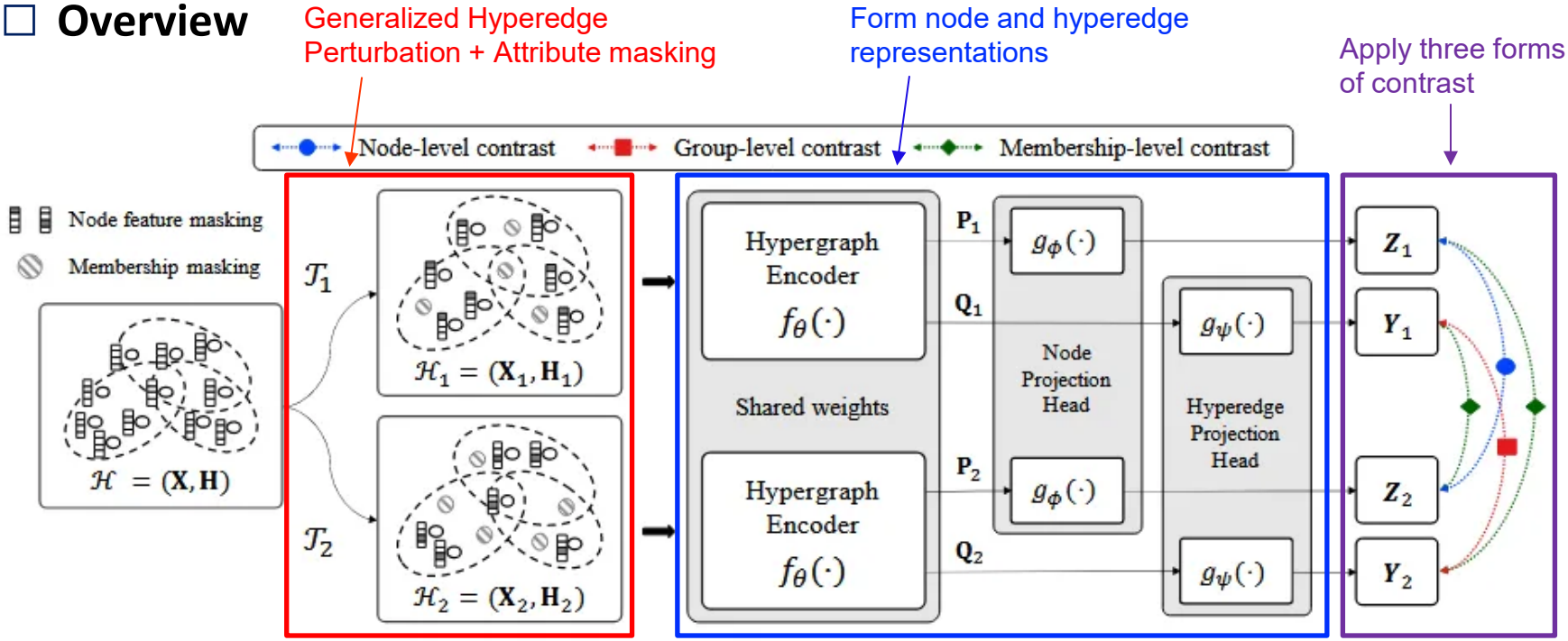
Tri-directional Contrast

- Node + Group + Membership contrast
- Preserve higher-order structural information
- Produce richer, more generalizable embeddings

TriCL

How to augment a hypergraph?	What to contrast?
Δ	O

☐ Overview



Lee, Dongjin, and Kijung Shin. "I'm me, we're us, and i'm us: Tri-directional contrastive learning on hypergraphs." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 37. No. 7. 2023.

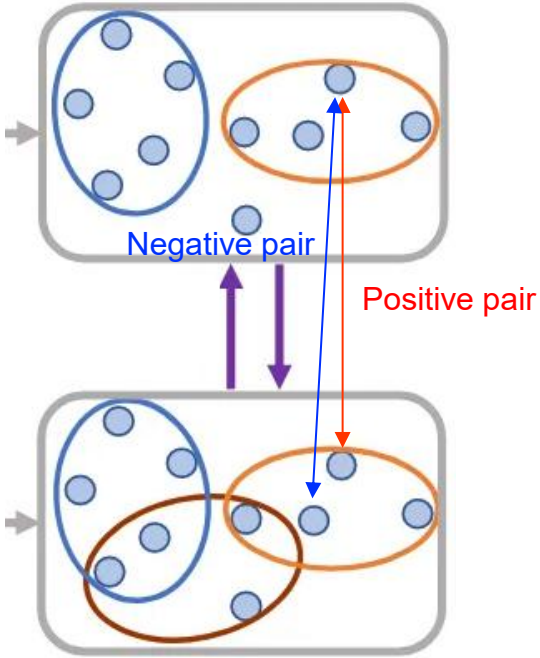
TriCL

Node-level contrast

- Discriminate the representations of the same node in the two augmented views from other node representations

$$\ell_n(\boldsymbol{z}_{1,i}, \boldsymbol{z}_{2,i}) = -\log \frac{e^{s(\boldsymbol{z}_{1,i}, \boldsymbol{z}_{2,i})/\tau_n}}{\sum_{k=1}^{|V|} e^{s(\boldsymbol{z}_{1,i}, \boldsymbol{z}_{2,k})/\tau_n}},$$

$$\mathcal{L}_n = \frac{1}{2|V|} \sum_{i=1}^{|V|} \left\{ \ell_n(\boldsymbol{z}_{1,i}, \boldsymbol{z}_{2,i}) + \ell_n(\boldsymbol{z}_{2,i}, \boldsymbol{z}_{1,i}) \right\}.$$



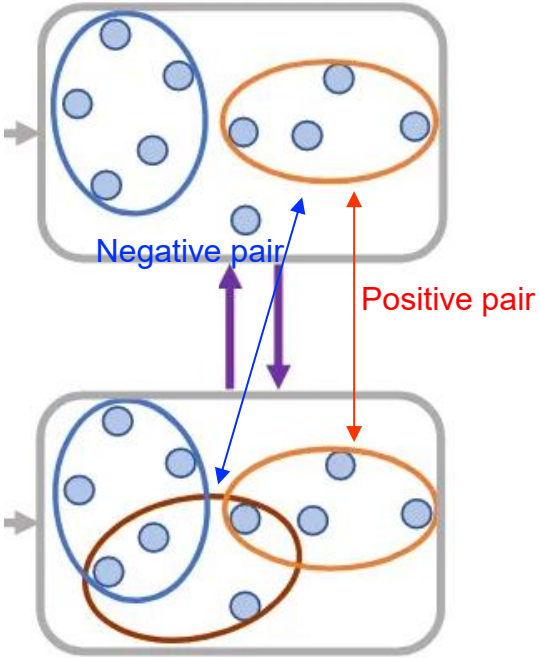
TriCL

Group-level contrast

- Distinguish the representations of the same hyperedge in the two augmented views from other hyperedge representations

$$\ell_g(\boldsymbol{y}_{1,j}, \boldsymbol{y}_{2,j}) = -\log \frac{e^{s(\boldsymbol{y}_{1,j}, \boldsymbol{y}_{2,j})/\tau_g}}{\sum_{k=1}^{|E|} e^{s(\boldsymbol{y}_{1,j}, \boldsymbol{y}_{2,k})/\tau_g}},$$

$$\mathcal{L}_g = \frac{1}{2|E|} \sum_{j=1}^{|E|} \{\ell_g(\boldsymbol{y}_{1,j}, \boldsymbol{y}_{2,j}) + \ell_g(\boldsymbol{y}_{2,j}, \boldsymbol{y}_{1,j})\}.$$



TriCL

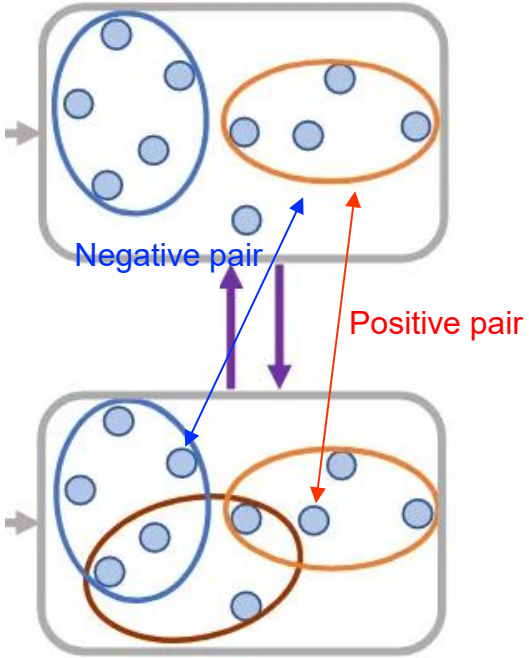
Membership-level contrast

- Learn to represent if relationships actually exist and to move away if they don't exist

$$\ell_m(\boldsymbol{z}_i, \boldsymbol{y}_j) = \underbrace{-\log \frac{e^{\mathcal{D}(\boldsymbol{z}_i, \boldsymbol{y}_j)/\tau_m}}{e^{\mathcal{D}(\boldsymbol{z}_i, \boldsymbol{y}_j)/\tau_m} + \sum_{k:i \notin k} e^{\mathcal{D}(\boldsymbol{z}_i, \boldsymbol{y}_k)/\tau_m}}}_{\text{when } \boldsymbol{z}_i \text{ is the anchor}}$$

$$\underbrace{-\log \frac{e^{\mathcal{D}(\boldsymbol{z}_i, \boldsymbol{y}_j)/\tau_m}}{e^{\mathcal{D}(\boldsymbol{z}_i, \boldsymbol{y}_j)/\tau_m} + \sum_{k:k \notin j} e^{\mathcal{D}(\boldsymbol{z}_k, \boldsymbol{y}_j)/\tau_m}}}_{\text{when } \boldsymbol{y}_j \text{ is the anchor}},$$

$$\mathcal{L}_m = \frac{1}{2K} \sum_{i=1}^{|V|} \sum_{j=1}^{|E|} \mathbb{1}_{[h_{ij}=1]} \left\{ \ell_m(\boldsymbol{z}_{1,i}, \boldsymbol{y}_{2,j}) + \ell_m(\boldsymbol{z}_{2,i}, \boldsymbol{y}_{1,j}) \right\}.$$



Experiments

Performance on Node Classification

- Graph contrastive learning methods show significantly lower accuracy compared to TriCL

Table 1: Node classification accuracy and standard deviations. Graph methods, marked as ⋆, are applied after converting hypergraphs to graphs via clique expansion. For each dataset, the best and the second-best performances are highlighted in **boldface** and underlined, respectively. A.R. denotes average rank, OOT denotes cases where results are not obtained within 24 hours, and OOM indicates out of memory on a 24GB GPU. In most cases, TriCL outperforms all others, including the supervised ones.

	Method	Cora-C	Citeseer	Pubmed	Cora-A	DBLP	Zoo	20News	Mushroom	NTU2012	ModelNet40	A.R.↓
Supervised	MLP	60.32 ± 1.5	62.06 ± 2.3	76.27 ± 1.1	64.05 ± 1.4	81.18 ± 0.2	75.62 ± 9.5	79.19 ± 0.5	99.58 ± 0.3	65.17 ± 2.3	93.75 ± 0.6	12.5
	GCN⋆	77.11 ± 1.8	66.07 ± 2.4	82.63 ± 0.6	73.66 ± 1.3	87.58 ± 0.2	36.79 ± 9.6	OOM	92.47 ± 0.9	71.17 ± 2.4	91.67 ± 0.2	11.7
	GAT⋆	77.75 ± 2.1	67.62 ± 2.5	81.96 ± 0.7	74.52 ± 1.3	88.59 ± 0.1	36.48 ± 10.0	OOM	OOM	70.94 ± 2.6	91.43 ± 0.3	11
	HGNN	77.50 ± 1.8	66.16 ± 2.3	83.52 ± 0.7	74.38 ± 1.2	88.32 ± 0.3	78.58 ± 11.1	80.15 ± 0.3	98.59 ± 0.5	72.03 ± 2.4	92.23 ± 0.2	8.1
	HyperConv	76.19 ± 2.1	64.12 ± 2.6	83.42 ± 0.6	73.52 ± 1.0	88.83 ± 0.2	62.53 ± 14.5	79.83 ± 0.4	97.56 ± 0.6	72.62 ± 2.6	91.84 ± 0.1	9.8
	HNHN	76.21 ± 1.7	67.28 ± 2.2	80.97 ± 0.9	74.88 ± 1.6	86.71 ± 1.2	78.89 ± 10.2	79.51 ± 0.4	99.78 ± 0.1	71.45 ± 3.2	92.96 ± 0.2	8.9
	HyperGCN	64.11 ± 7.4	59.92 ± 9.6	78.40 ± 9.2	60.65 ± 9.2	76.59 ± 7.6	40.86 ± 2.1	77.31 ± 6.0	48.26 ± 0.3	46.05 ± 3.9	69.23 ± 2.8	15.1
	HyperSAGE	64.98 ± 5.3	52.43 ± 9.4	79.49 ± 8.7	64.59 ± 4.3	79.63 ± 8.6	40.86 ± 2.1	OOT	OOT	OOT	OOT	14.7
	UniGCN	77.91 ± 1.9	66.40 ± 1.9	84.08 ± 0.7	77.30 ± 1.4	90.31 ± 0.2	72.10 ± 12.1	**80.24 ± 0.4**	98.84 ± 0.5	73.27 ± 2.7	94.62 ± 0.2	5.9
	AllSet	76.21 ± 1.7	67.83 ± 1.8	82.85 ± 0.9	76.94 ± 1.3	90.07 ± 0.3	72.72 ± 11.8	79.90 ± 0.4	99.78 ± 0.1	75.09 ± 2.5	96.85 ± 0.2	6.2
Unsupervised	Node2vec⋆	70.99 ± 1.4	53.85 ± 1.9	78.75 ± 0.9	58.50 ± 2.1	72.09 ± 0.3	17.02 ± 4.1	63.35 ± 1.7	88.16 ± 0.8	67.72 ± 2.1	84.94 ± 0.4	15.6
	DGI⋆	78.17 ± 1.4	68.81 ± 1.8	80.83 ± 0.6	76.94 ± 1.1	88.00 ± 0.2	36.54 ± 9.7	OOM	OOM	72.01 ± 2.5	92.18 ± 0.2	9.3
	GRACE⋆	79.11 ± 1.7	68.65 ± 1.7	80.08 ± 0.7	76.59 ± 1.0	OOM	37.07 ± 9.3	OOM	OOM	70.51 ± 2.4	90.68 ± 0.3	10.4
	S^2-HHGR	78.08 ± 1.7	68.21 ± 1.8	82.13 ± 0.6	78.15 ± 1.1	88.69 ± 0.2	80.06 ± 11.1	79.75 ± 0.3	97.15 ± 0.5	73.95 ± 2.4	93.26 ± 0.2	6.8
	Random-Init	63.62 ± 3.1	60.44 ± 2.5	67.49 ± 2.2	66.27 ± 2.2	76.57 ± 0.6	78.43 ± 11.0	77.14 ± 0.6	97.40 ± 0.6	74.39 ± 2.6	96.29 ± 0.3	11.9
	TriCL-N	80.23 ± 1.2	70.28 ± 1.5	83.44 ± 0.6	81.94 ± 1.1	90.88 ± 0.1	79.94 ± 11.1	80.18 ± 0.2	99.76 ± 0.2	75.20 ± 2.6	97.01 ± 0.2	3.4
	TriCL-NG	81.45 ± 1.2	71.38 ± 1.2	83.68 ± 0.7	82.00 ± 1.0	90.94 ± 0.1	80.19 ± 11.1	80.18 ± 0.2	99.81 ± 0.1	**75.25 ± 2.5**	97.02 ± 0.1	2
	TriCL	**81.57 ± 1.1**	**72.02 ± 1.2**	**84.26 ± 0.6**	**82.15 ± 0.9**	**91.12 ± 0.1**	**80.25 ± 11.2**	80.14 ± 0.2	**99.83 ± 0.1**	75.23 ± 2.4	**97.08 ± 0.1**	**1.5**

Lee, Dongjin, and Kijung Shin. "I'm me, we're us, and i'm us: Tri-directional contrastive learning on hypergraphs." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 37. No. 7. 2023.

Experiments

☐ Ablation Study

- Considering the different types of contrast together can help improve performance

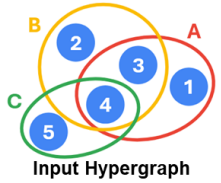
Table 2: Comparison of node classification accuracy according to whether or not to use each type of contrast (i.e., $\mathcal{L}_n$, $\mathcal{L}_g$, and $\mathcal{L}_m$). Using all types of contrasts (i.e., node-, group-, and membership-level contrast) achieves the best performance in most cases as they are complementarily reinforcing each other.

$\mathcal{L}_n$	$\mathcal{L}_g$	$\mathcal{L}_m$	Cora-C	Citeseer	Pubmed	Cora-A	DBLP	Zoo	20News	Mushroom	NTU2012	ModelNet40	A.R.↓
✓	-	-	80.23 ± 1.2	70.28 ± 1.5*	83.44 ± 0.6	81.94 ± 1.1	90.88 ± 0.1	79.94 ± 11.1	**80.18 ± 0.2**	99.76 ± 0.2	75.20 ± 2.6	97.01 ± 0.2	3.8
-	✓	-	79.69 ± 1.6	71.02 ± 1.3*	80.20 ± 1.3	78.98 ± 1.4	88.60 ± 0.2	79.31 ± 10.7	79.35 ± 0.4	99.13 ± 0.3	74.41 ± 2.6	96.66 ± 0.2	5.7
-	-	✓	76.76 ± 1.8	63.98 ± 2.0	79.86 ± 0.9	76.77 ± 1.1	63.95 ± 7.2	79.80 ± 11.0	79.27 ± 0.3	94.87 ± 0.7	73.11 ± 2.8	96.57 ± 0.2	6.9
✓	✓	-	81.45 ± 1.2	71.38 ± 1.4	83.68 ± 0.7	82.00 ± 1.0	90.94 ± 0.1	80.19 ± 11.1	**80.18 ± 0.2**	99.81 ± 0.1	**75.25 ± 2.5**	97.02 ± 0.1	2.3
✓	-	✓	80.49 ± 1.3	70.46 ± 1.5	83.98 ± 0.7	81.62 ± 1.0	90.75 ± 0.1	80.19 ± 11.1	80.15 ± 0.2	99.74 ± 0.2	75.12 ± 2.5	97.03 ± 0.1	3.6
-	✓	✓	80.80 ± 1.1	71.73 ± 1.4	82.81 ± 0.7	80.24 ± 1.0	90.17 ± 0.1	80.20 ± 11.1	79.29 ± 0.2	99.82 ± 0.1	73.76 ± 2.5	96.74 ± 0.1	4.1
✓	✓	✓	**81.57 ± 1.1**	**72.02 ± 1.4**	**84.26 ± 0.6**	**82.15 ± 0.9**	**91.12 ± 0.1**	**80.25 ± 11.2**	80.14 ± 0.2	**99.83 ± 0.1**	75.23 ± 2.4	**97.08 ± 0.1**	**1.4**

Lee, Dongjin, and Kijung Shin. "I'm me, we're us, and i'm us: Tri-directional contrastive learning on hypergraphs." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 37. No. 7. 2023.

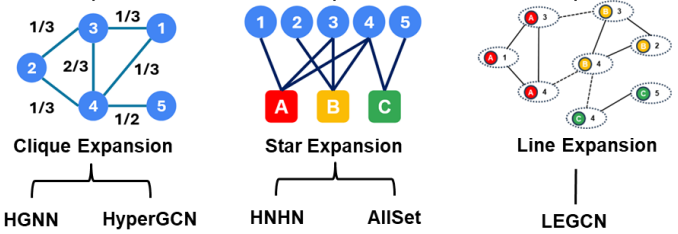
Conclusion

- **Hypergraphs Model Higher-Order Interactions**



Input Hypergraph

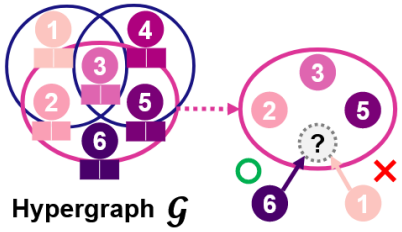
- **Hypergraph Expansions for Message Passing Design**
  - Clique Expansion: Vertex-to-Vertex
  - Star Expansion: Vertex-to-Hyperedge
  - Line Expansion: Incidence-to-Incidence



Clique Expansion | Star Expansion | Line Expansion

HGNN | HyperGCN | HNHN | AllSet | LEGCN

- **Self-Supervised Hypergraph Learning**
  - Generation-based Methods
  - Contrast-based Methods



Hypergraph $\mathcal{G}$